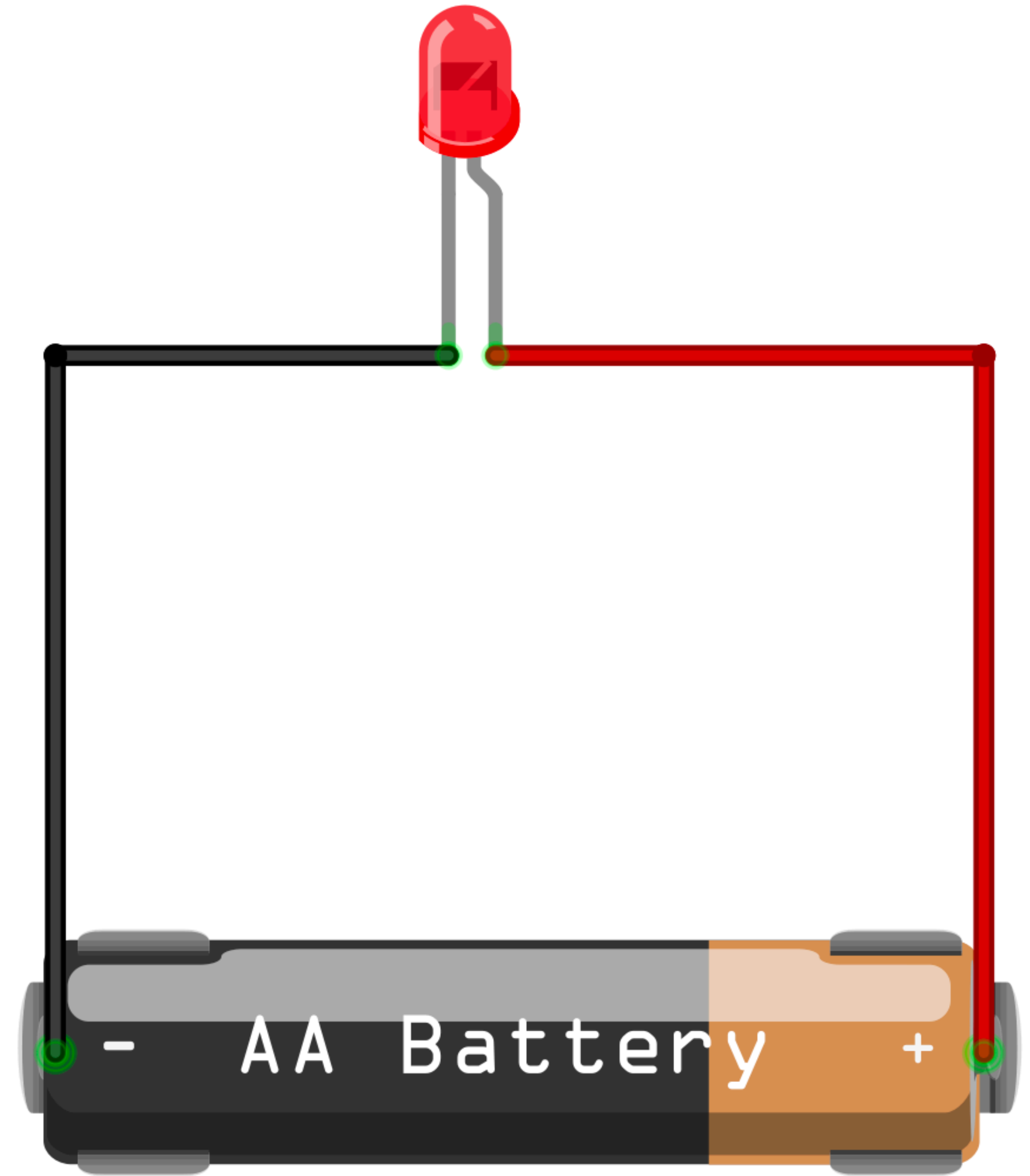


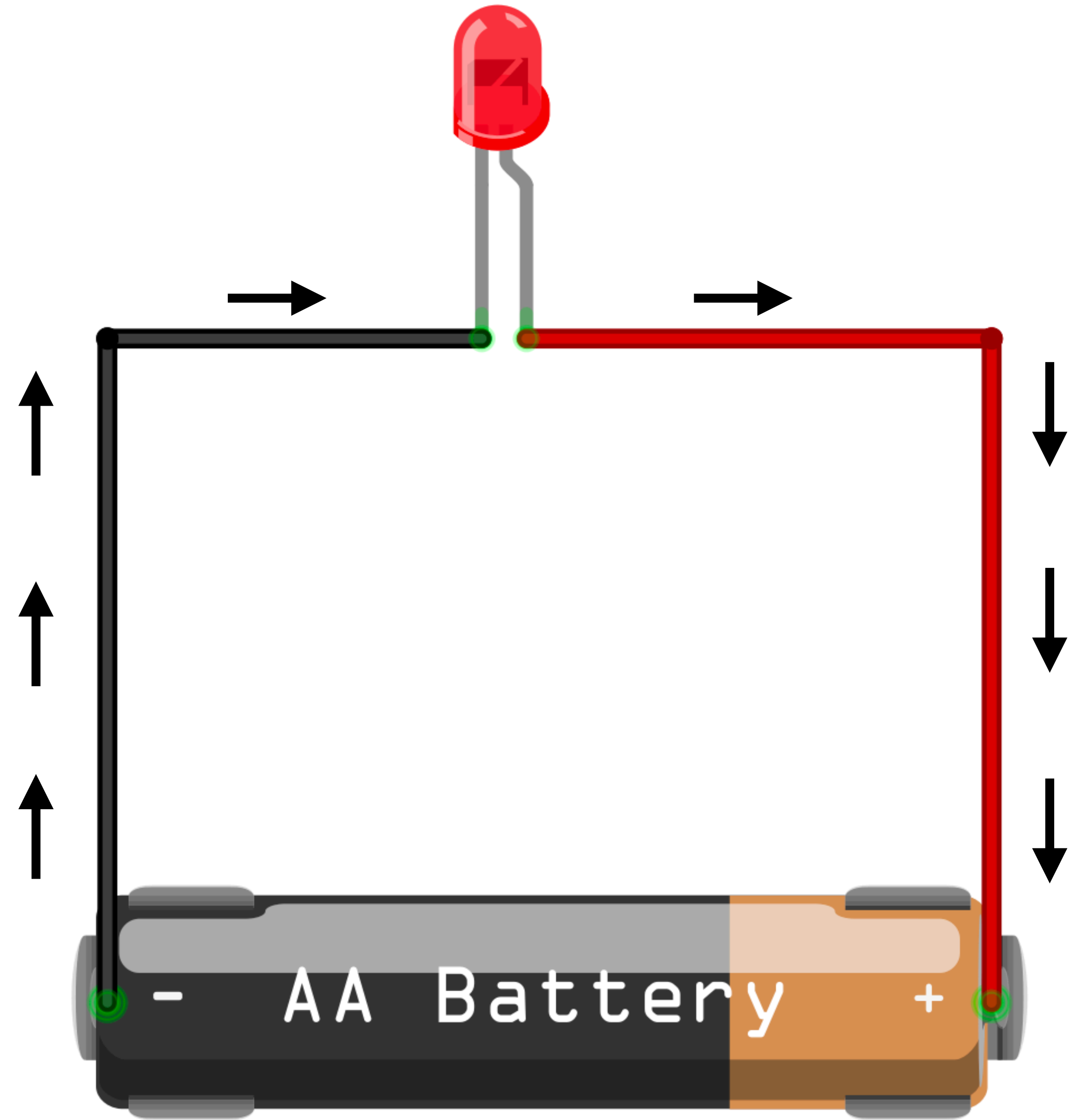
INTRODUCTION TO CIRCUITS AND ELECTRICITY (and the lab)

What is a Circuit?

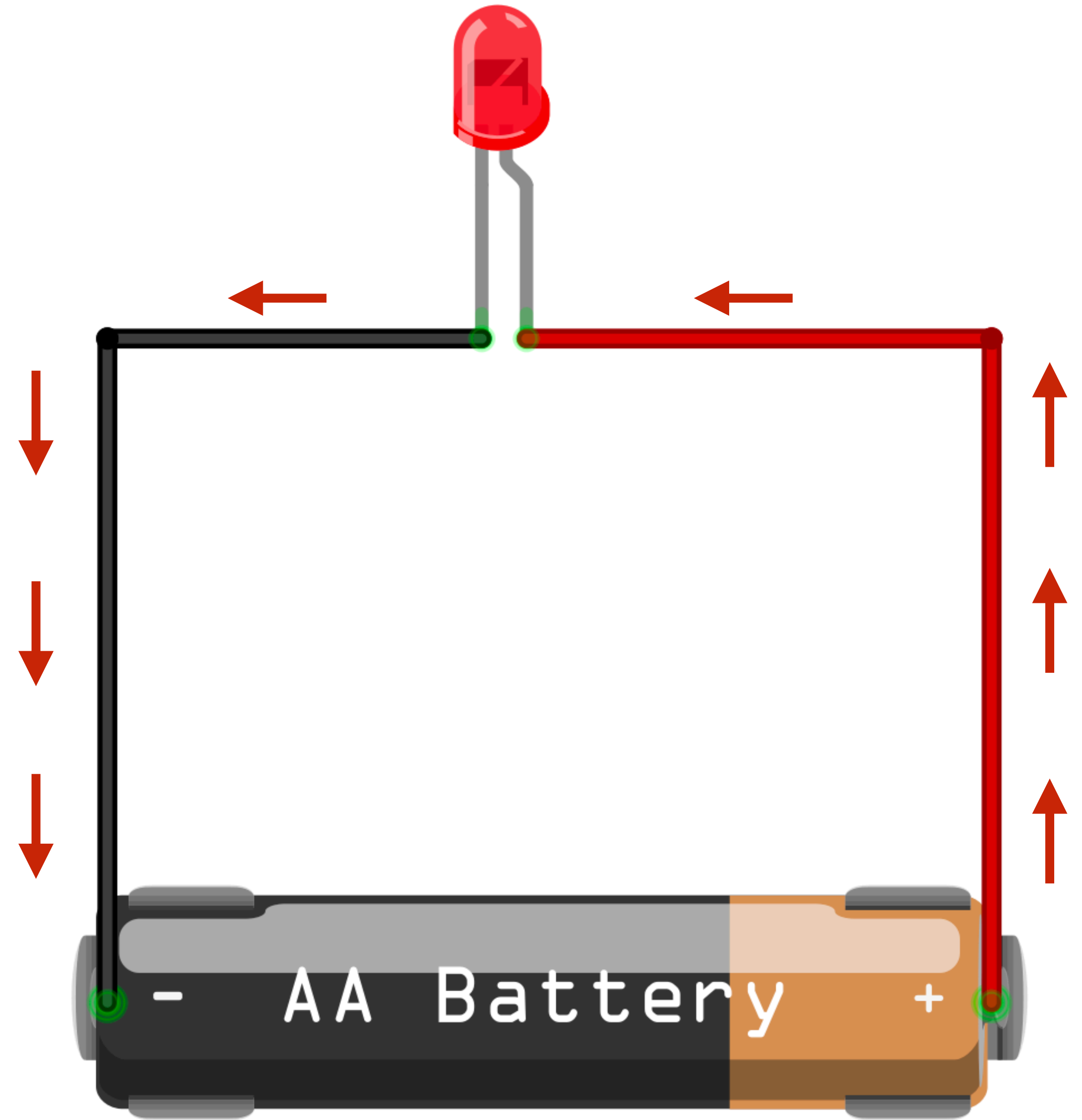
A circuit is a closed loop that carries a current.

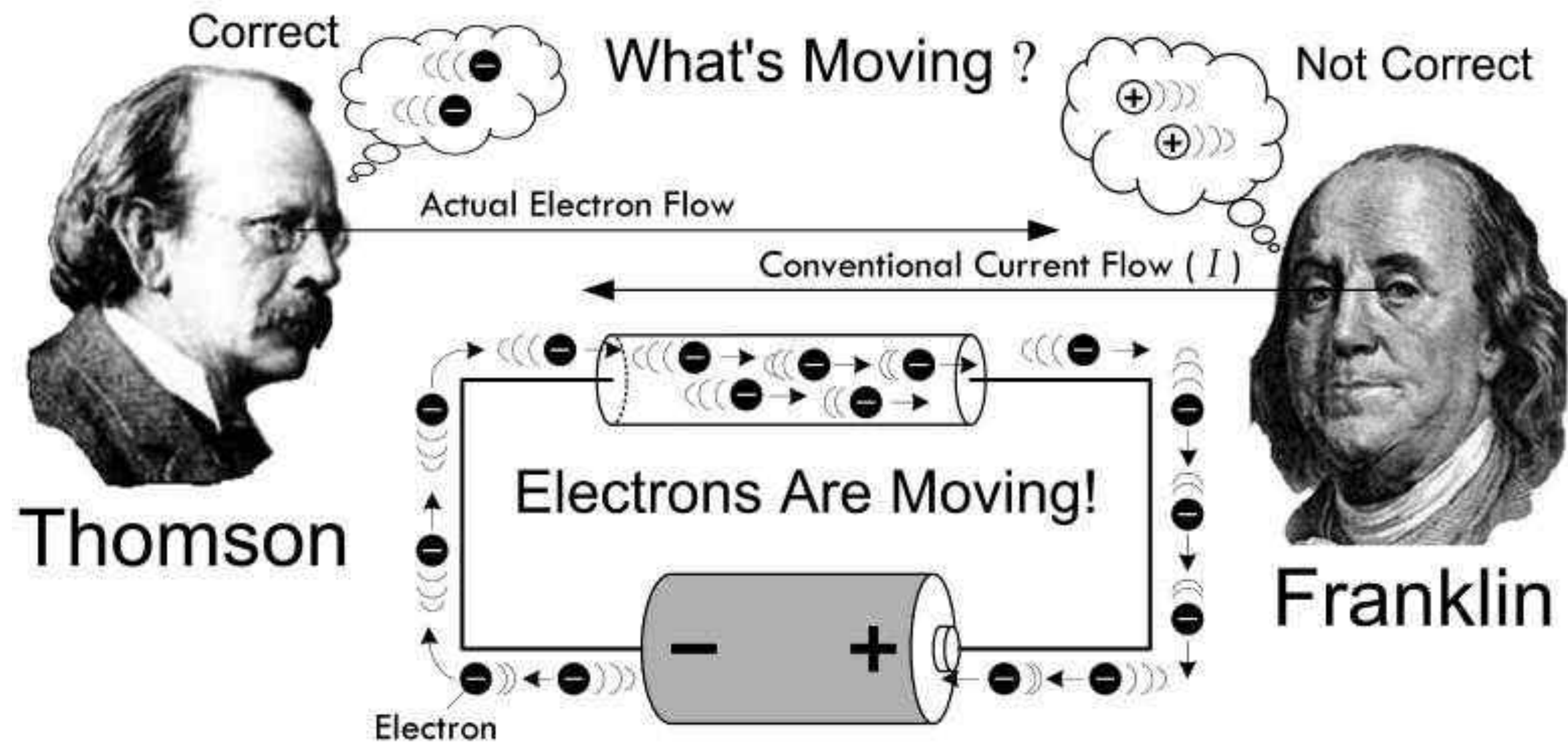


**The “actual”
direction of
current is from
negative to
positive.**



**But we
generally talk
as if the
current flows
the other
direction.**





DC - constant current flow in one direction. Think batteries, computer chargers, etc.

AC - Alternating current moves back-and-forth. This is what comes out of a wall socket.

(We mostly care about DC)

Three Key Elements of Electricity

1. Current
2. Voltage
3. Resistance

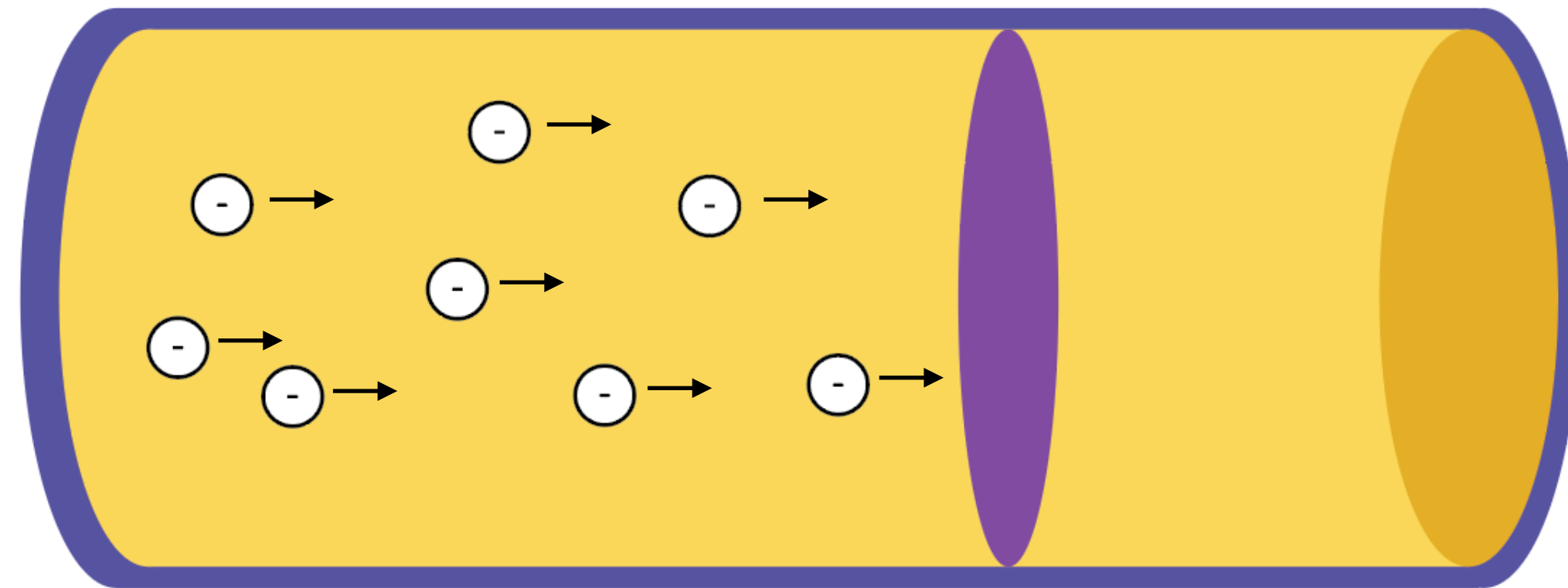
What is Current?

Current (I) is the measure of the flow of electricity through a point in the circuit.

Its unit — the ampere (A) — is defined as the travel of about 6.24×10^{18} elementary charges per second through a particular spot.

The “elementary charge” that’s moving in electronic circuits is typically the electron.

An amp is a measure of how many electrons pass through a point in a given amount of time.



What is Voltage?

Voltage—Measured in Volts and symbolized as “V”—is the “measure of electromotive force.” It is the “push” that causes the electrons to flow.

Voltage always refers to a relative measurement between two spots; if one of the measurement endpoints is not specified, the reference is usually the circuit’s negative terminal aka “ground”.

What is Resistance?

Resistance is how much a circuit resists the flow of current.

Measured in Ohms and symbolized with “ Ω ”

Ohm's Law

Ohm's law states that the current through a conductor is directly proportional to the voltage across the two points.

$$V = I \times R$$

V is voltage

I is current

R is resistance

$$V = I \times R$$

$$I = V / R$$

$$R = V / I$$

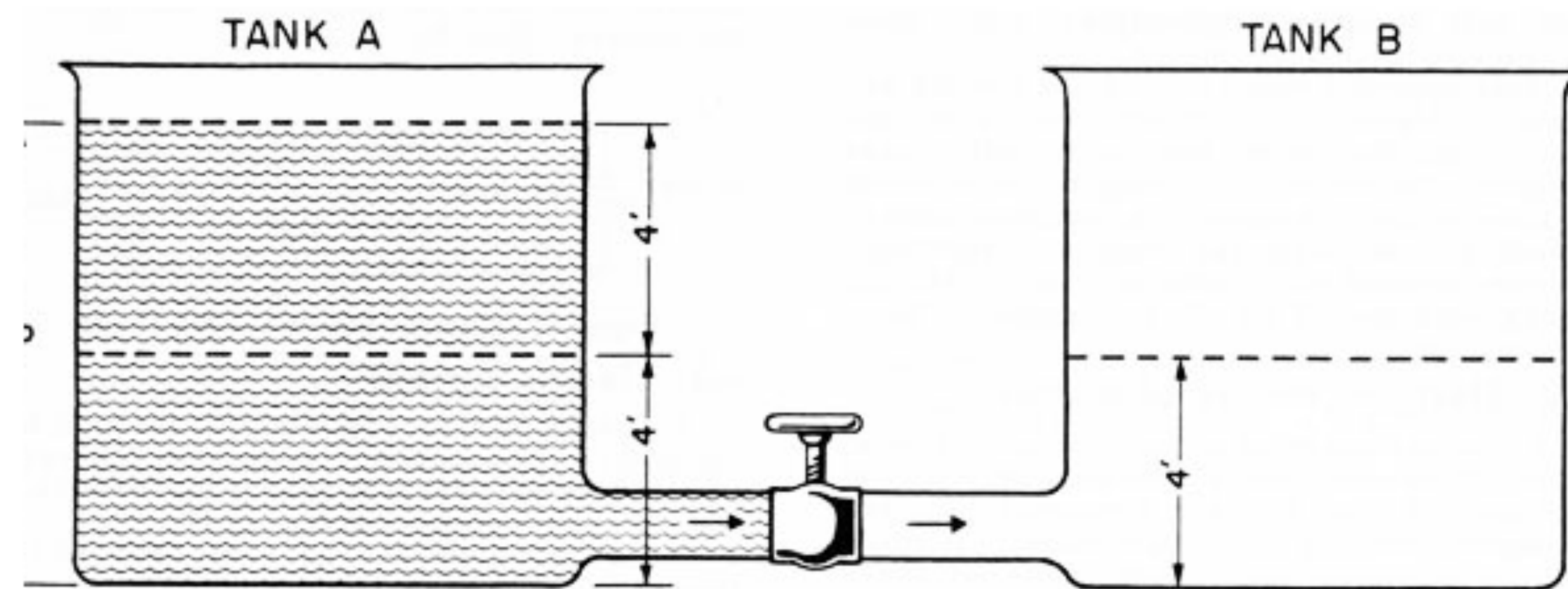
V is voltage

I is current

R is resistance

$$V = I \times R$$

This means that an increase in voltage causes a proportional increase in current.



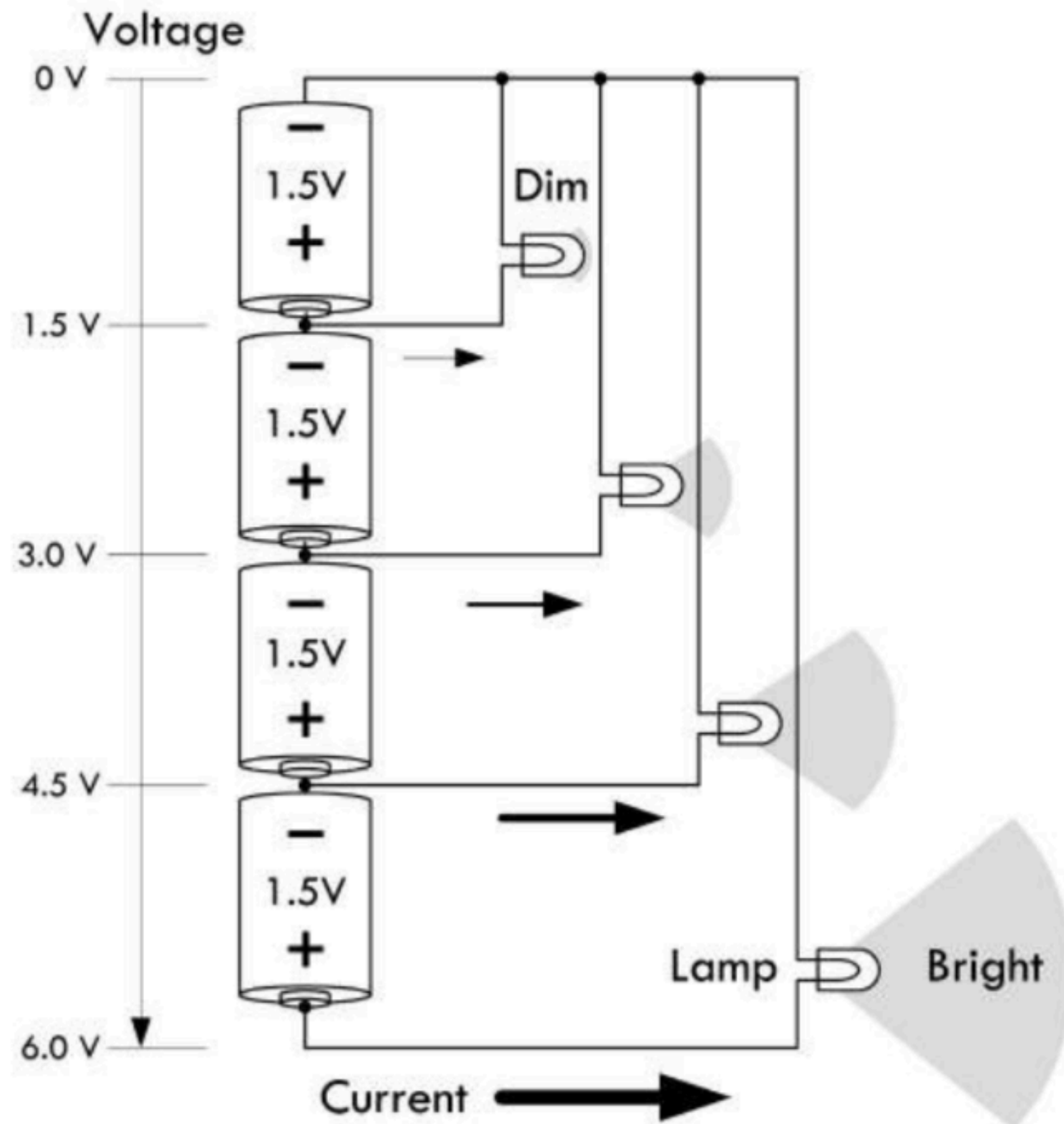
$$R \text{ (resistance)} = V \text{ (voltage)} / I \text{ (current)}$$

$$V = R * I$$

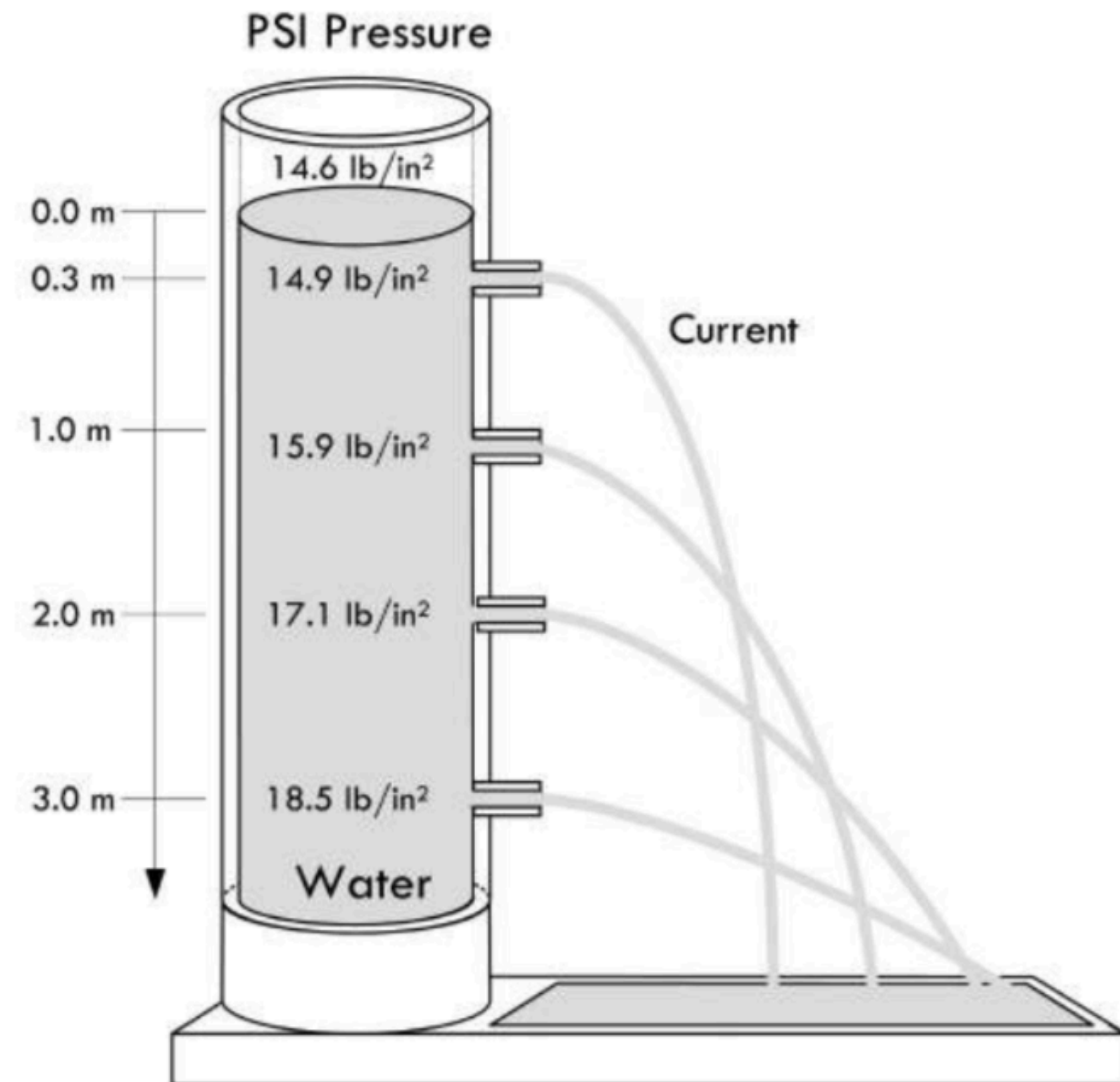
$$I = V / R$$

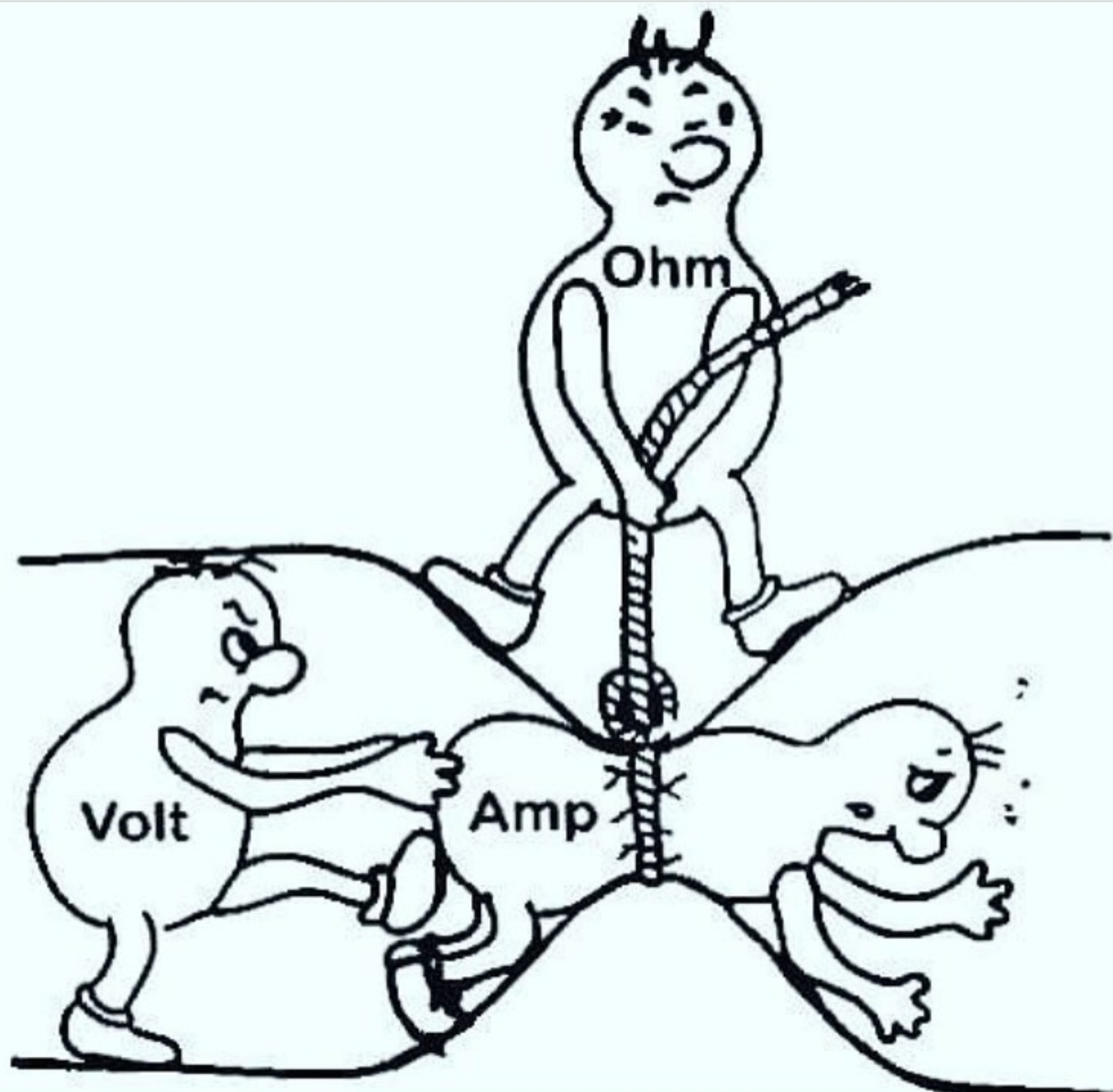
This is the only rule that you really have to memorise and learn to use, because in most of your work, this is the only one that you will really need.

Electrical System

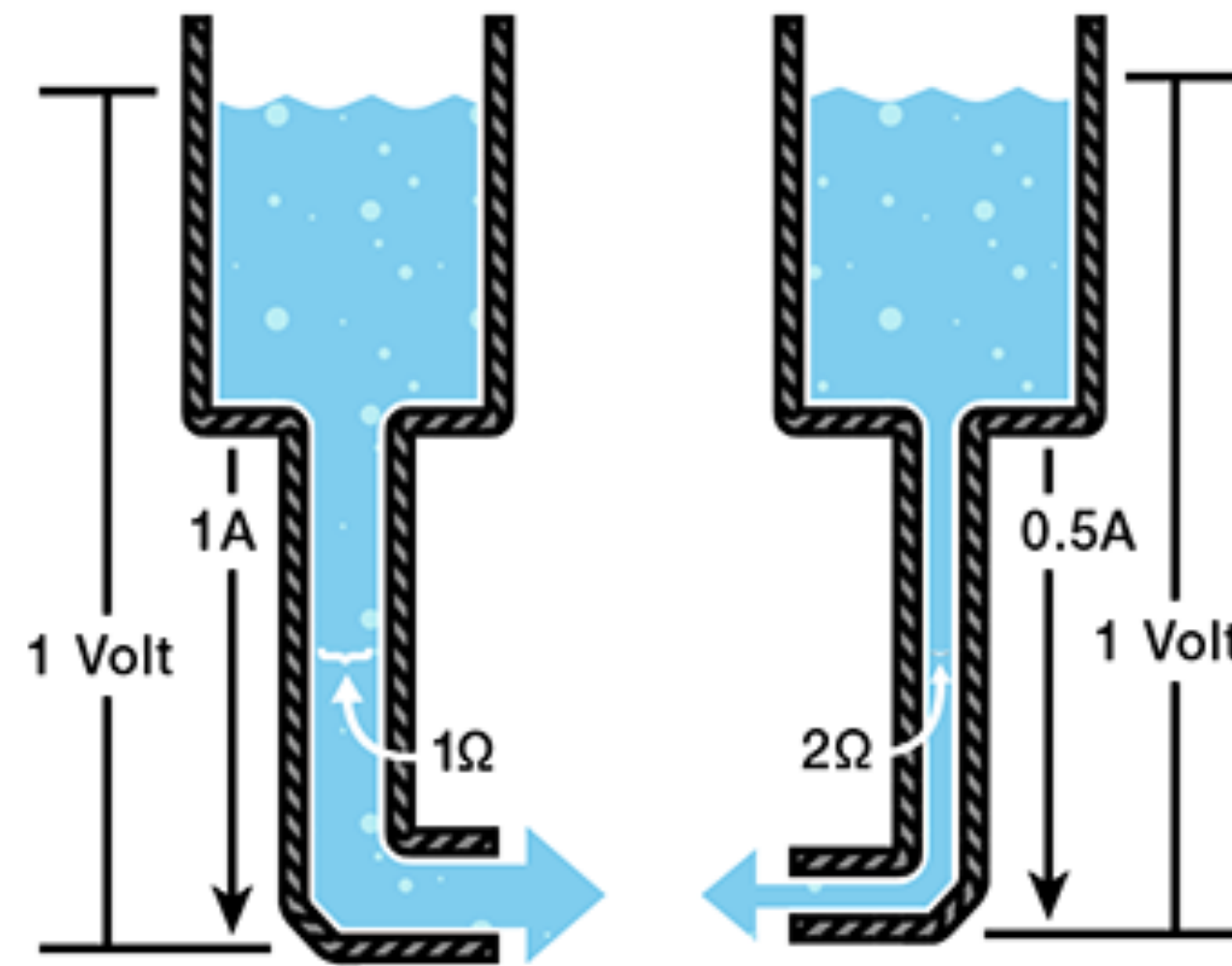


Water System





$$V = I \cdot R$$



V = Voltage in volts

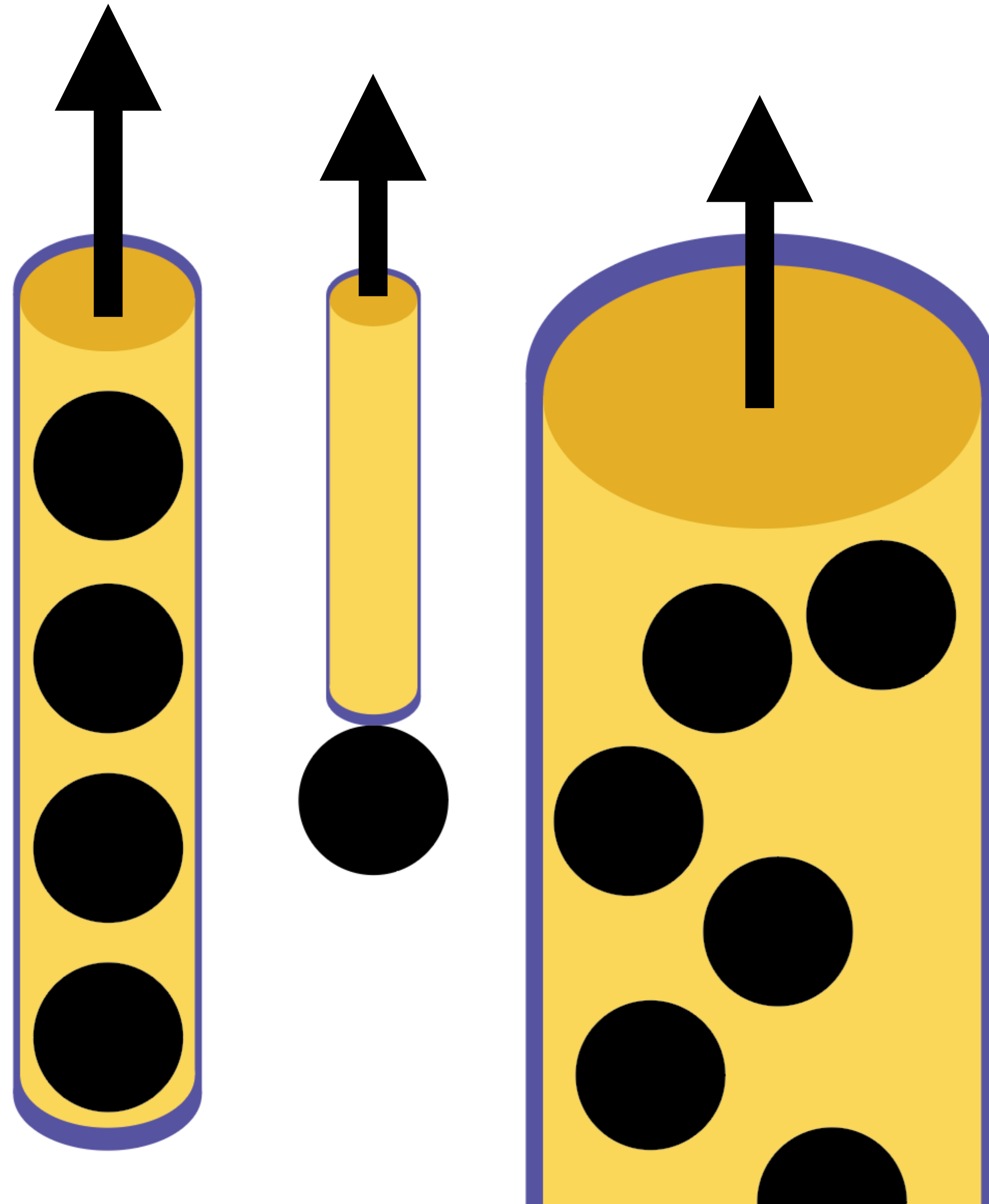
I = Current in amps

R = Resistance in ohms



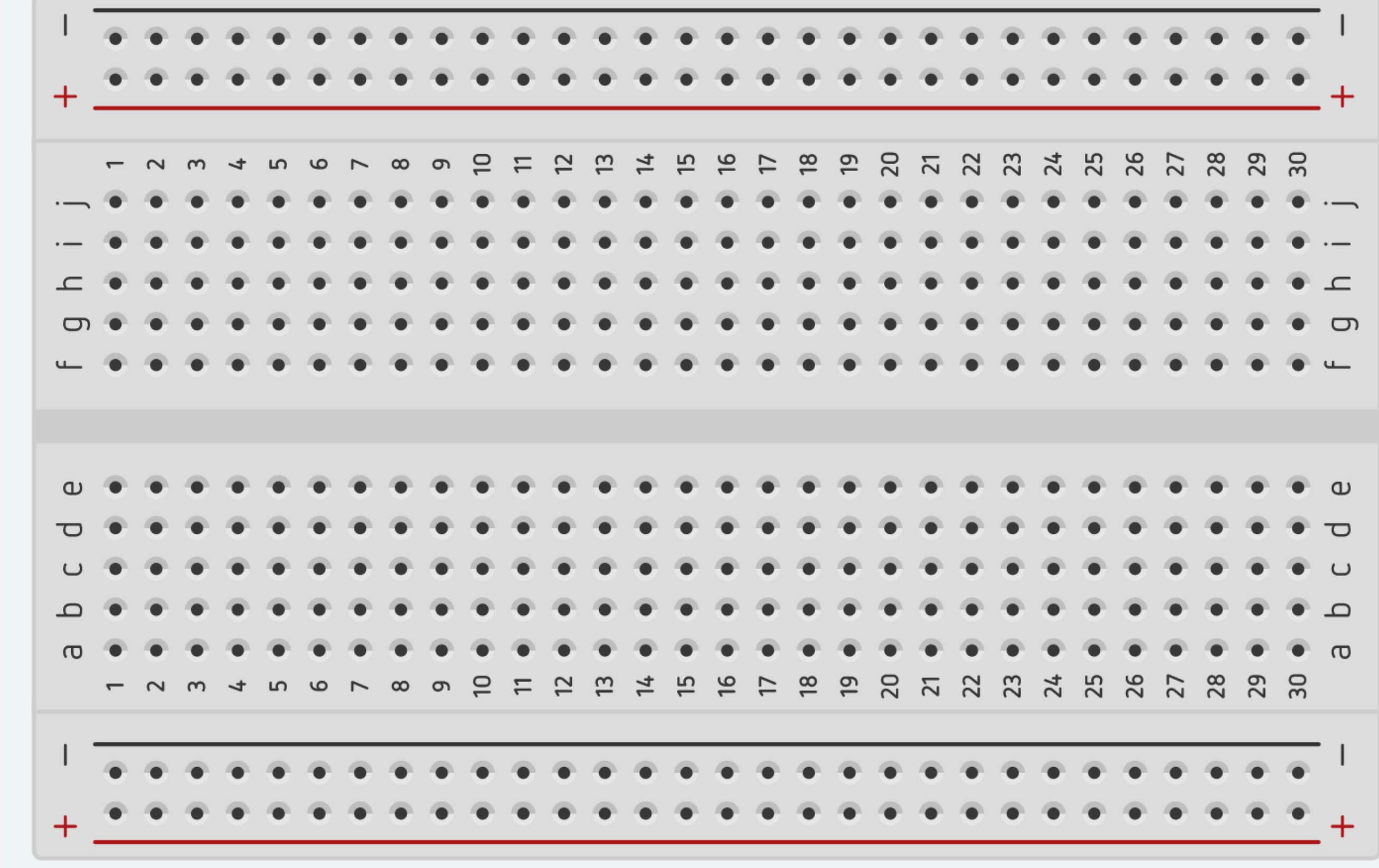
In the boba analogy,

- the straw size is the resistance,
- the strength of the sucking is the voltage
- and the number of boba per second is the current



ELECTRICAL COMPONENTS

Breadboard



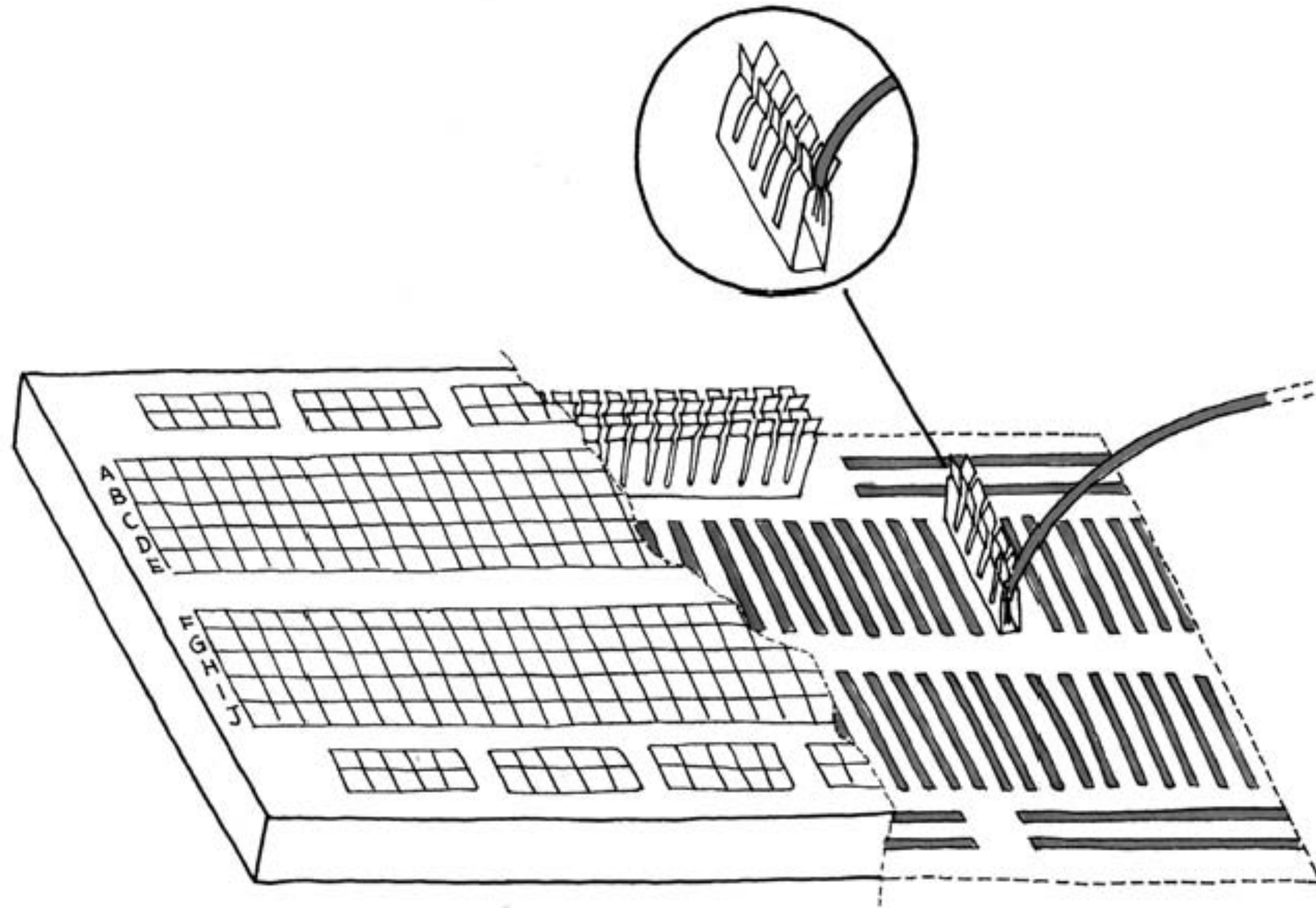
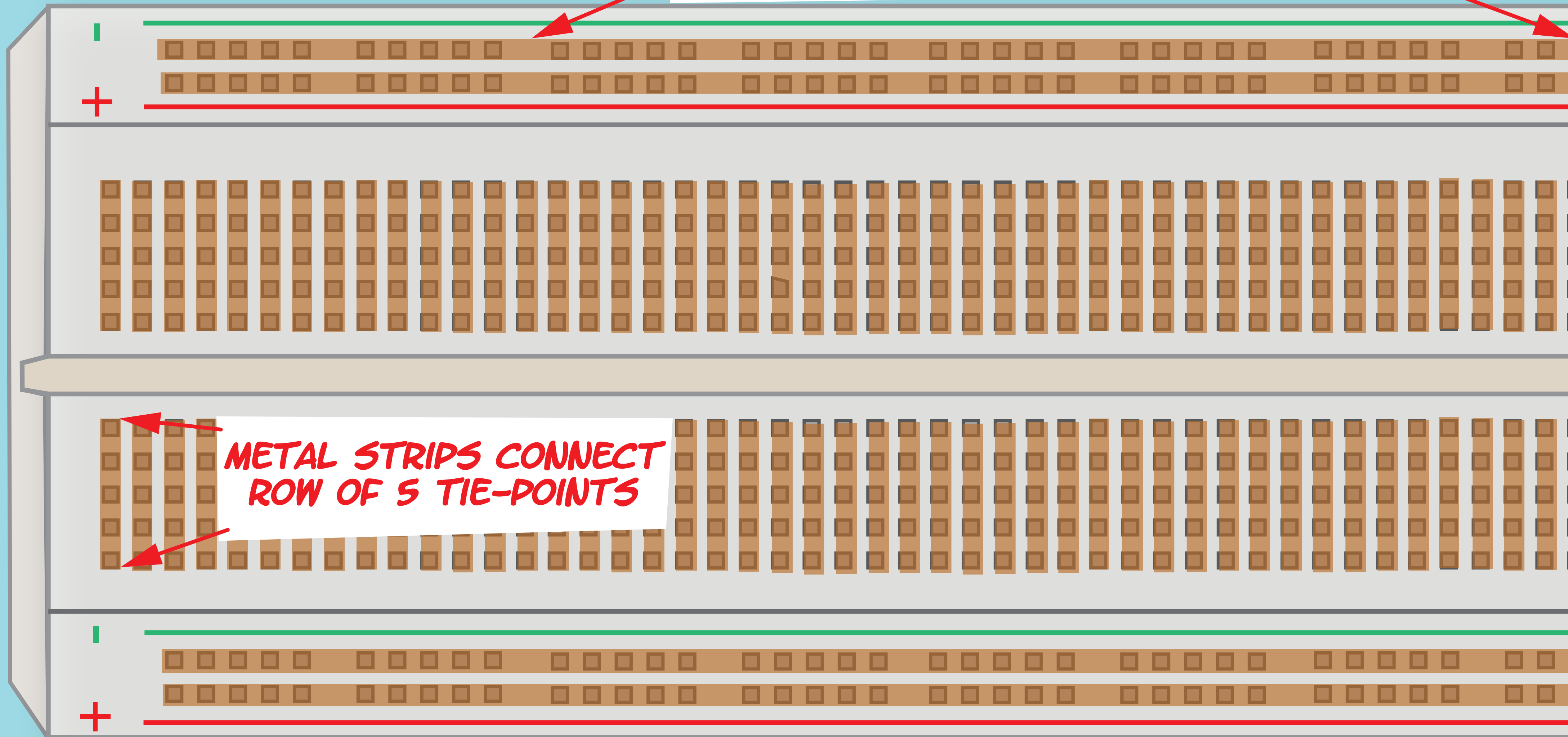
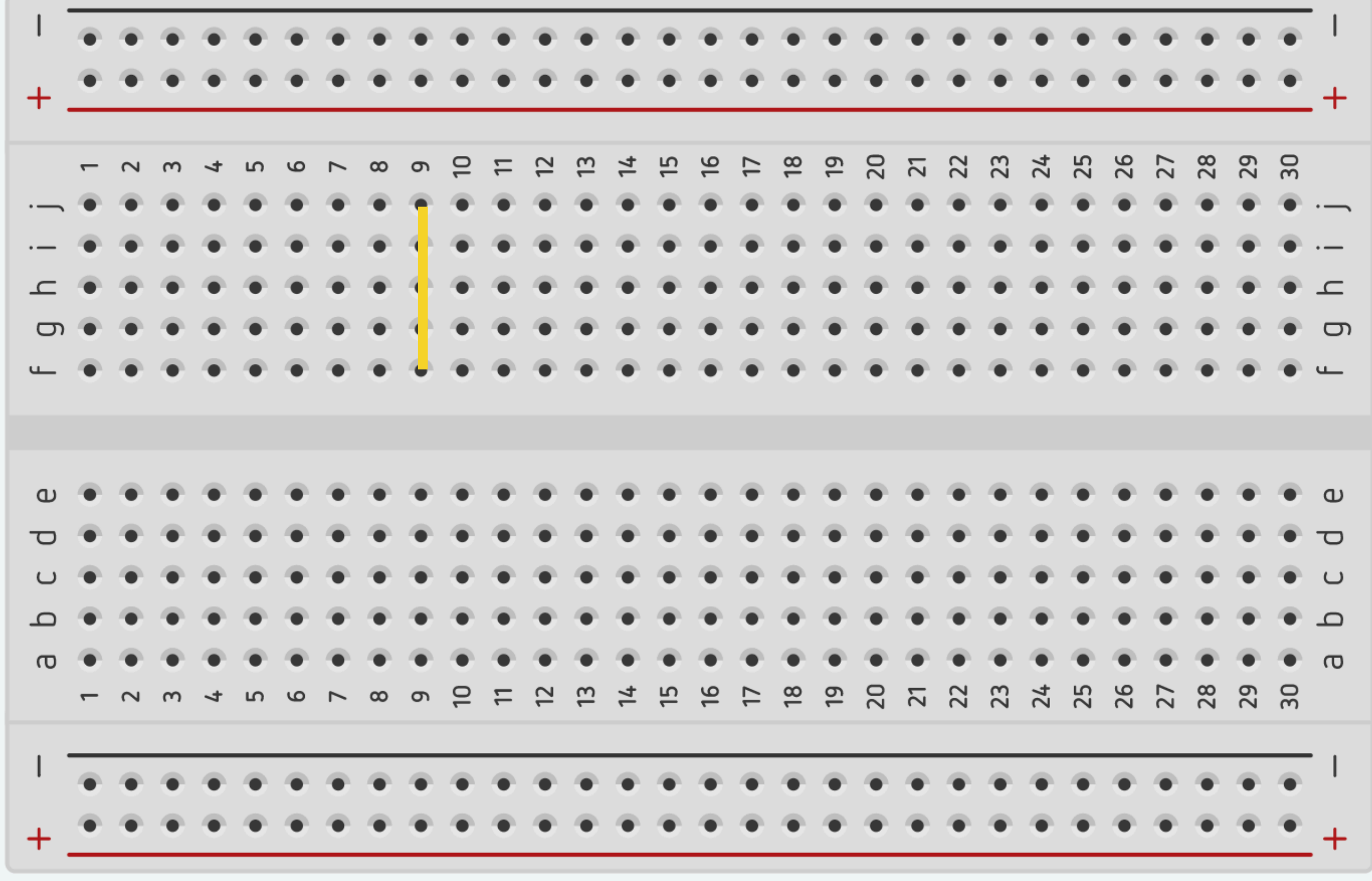


Figure A-1.
The solderless breadboard

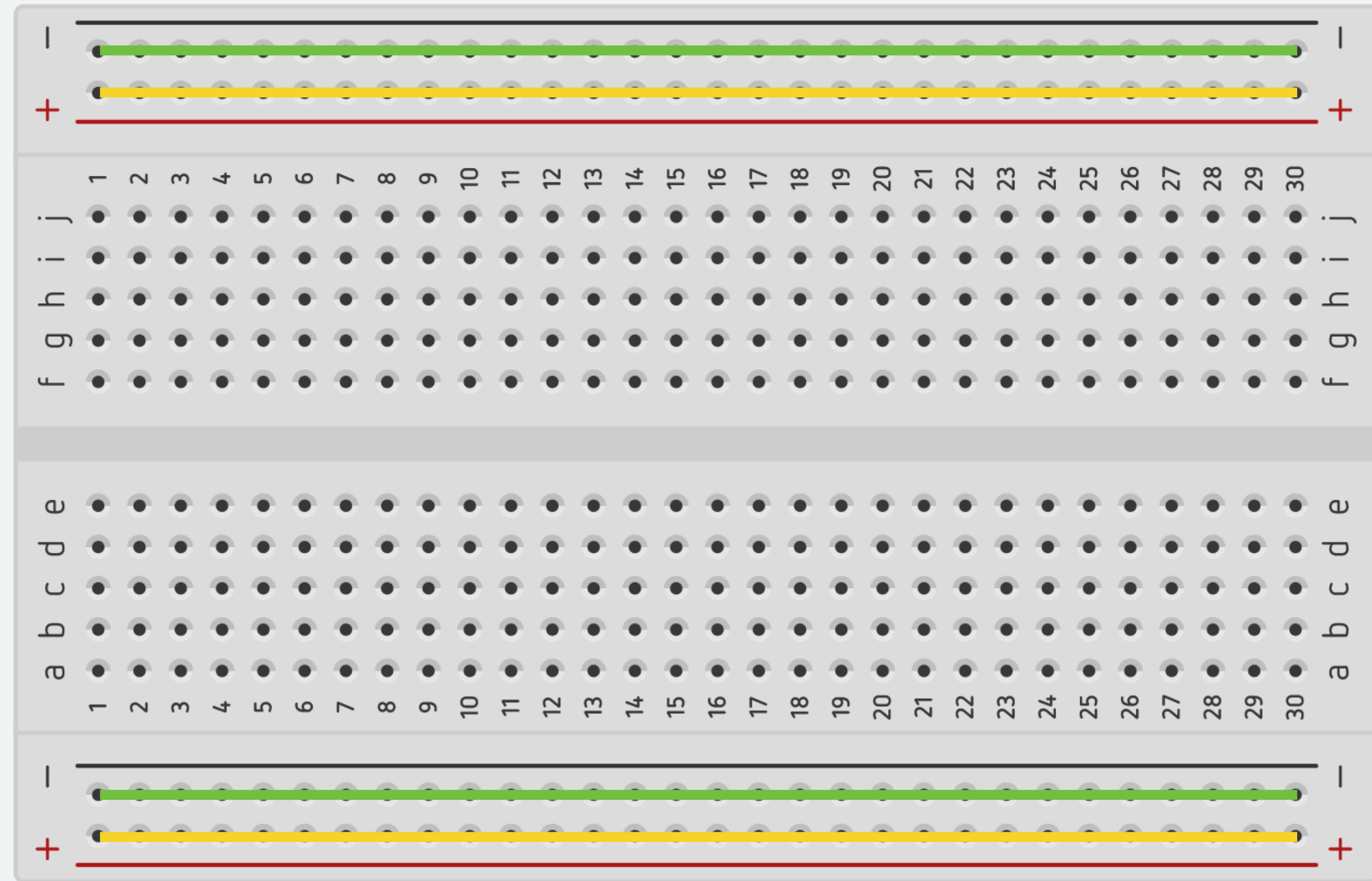
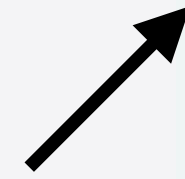
STRIPS OF METAL CONNECT
TIE-POINTS ON DISTRIBUTION BUS



METAL STRIPS CONNECT
ROW OF 5 TIE-POINTS

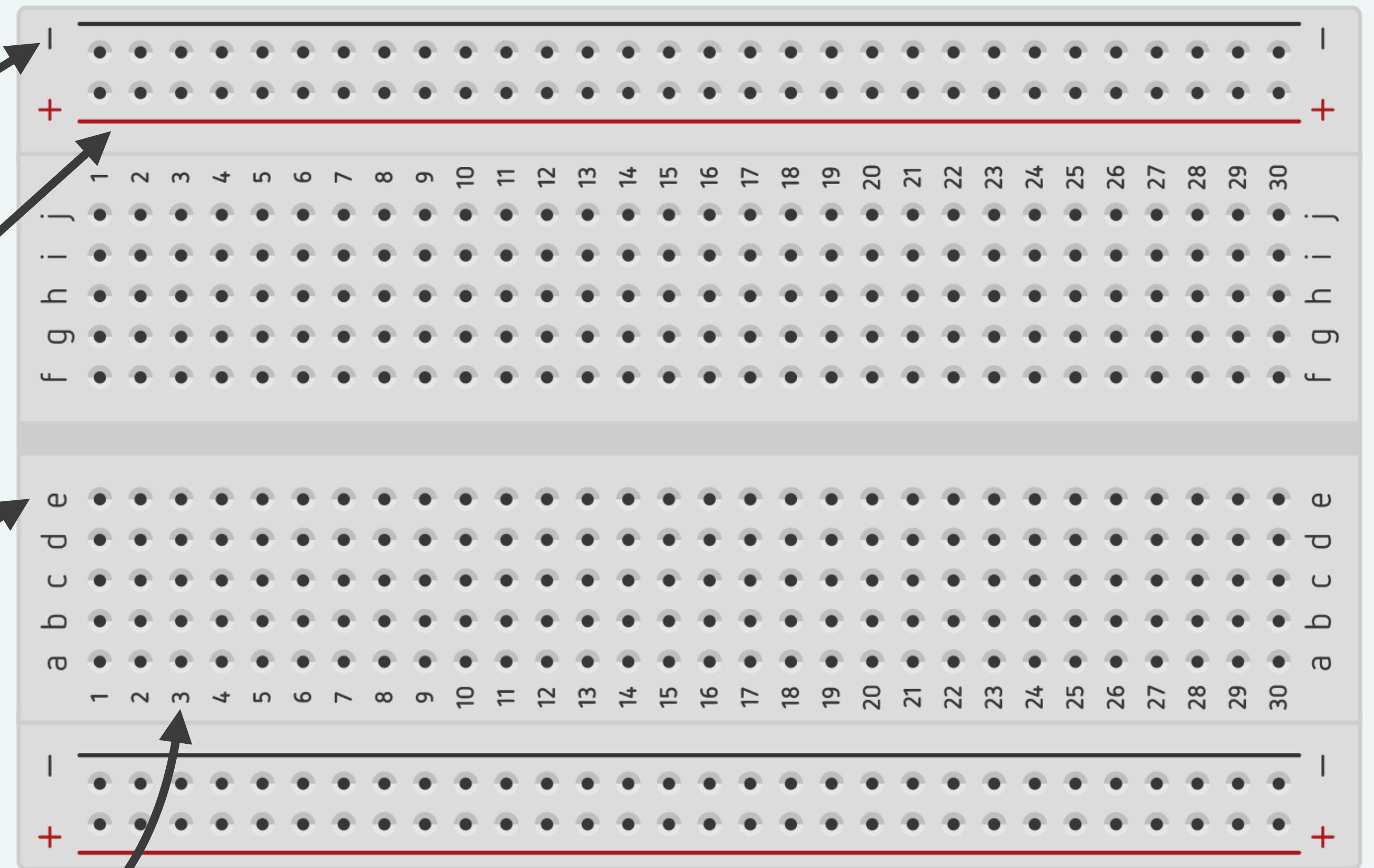


RAILS
(typically for
connecting
power)



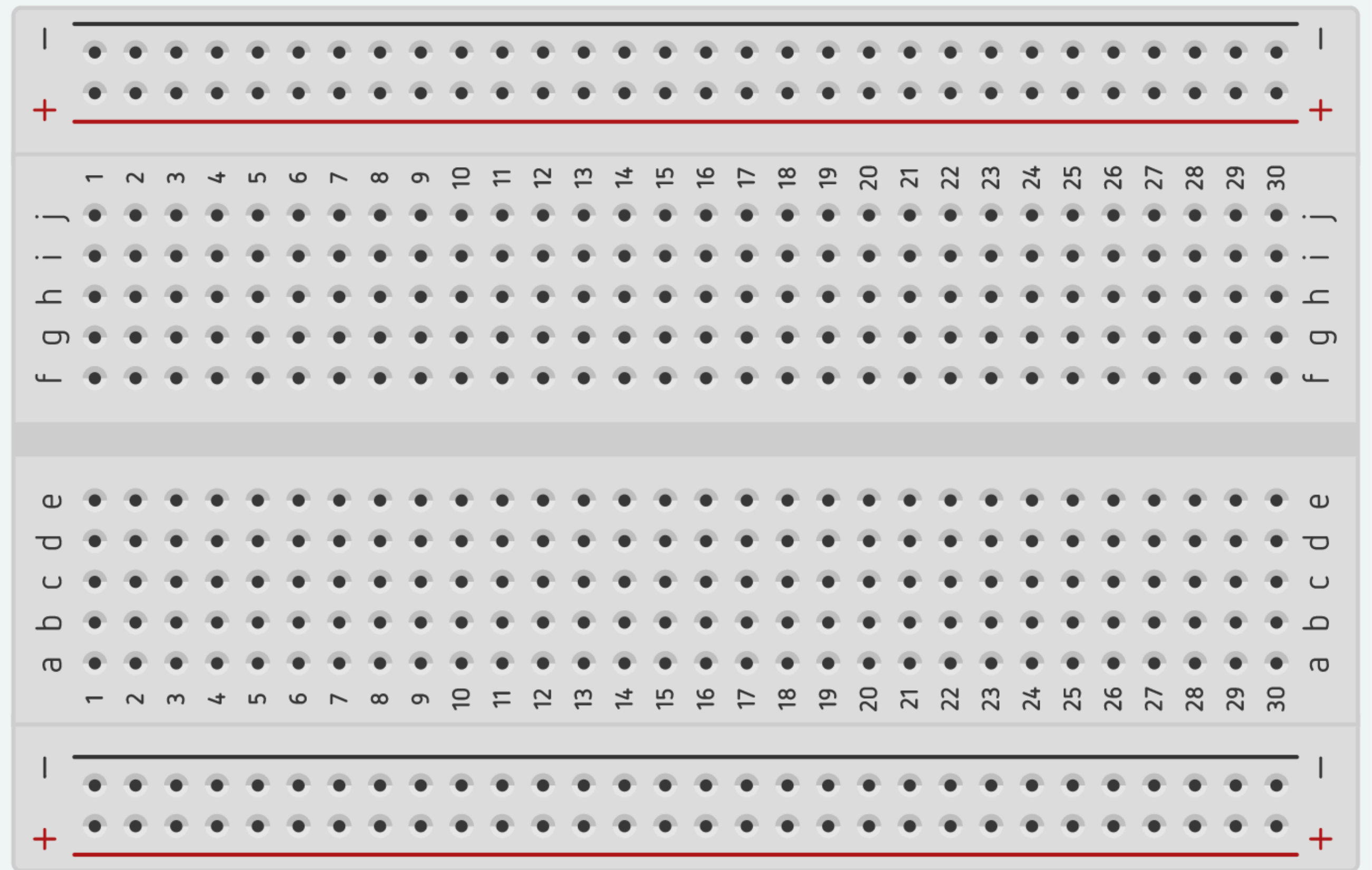
Different boards have different markings.

Some have symbols
others have colors
and most have rows
labeled with letters,
and columns labeled
with numbers.



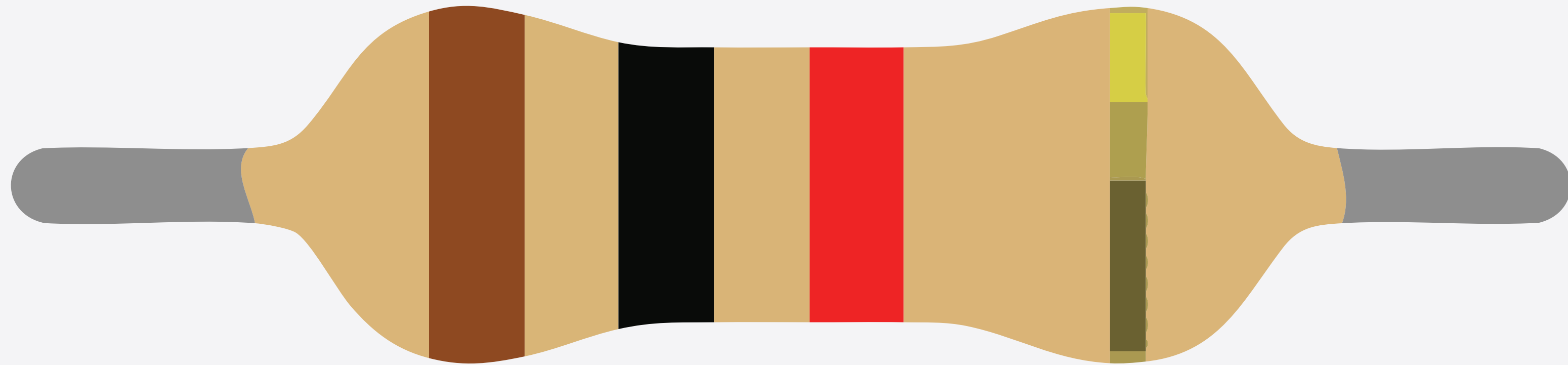
These don't really mean anything for the circuit itself, they are just there for you, sorta like a comment when you are coding.

I try to keep ground connected to black and power connected to red.



Resistors

Resistors



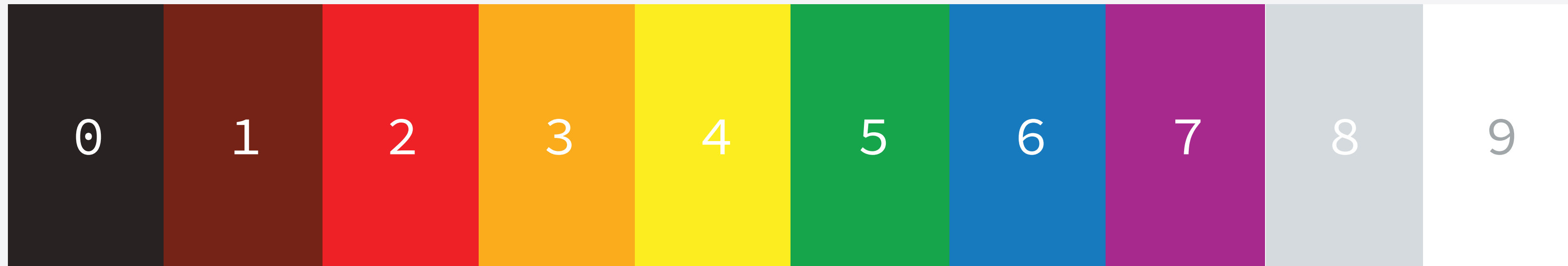
A resistor is a
component that **creates
electrical resistance** in
a circuit.

Resistors



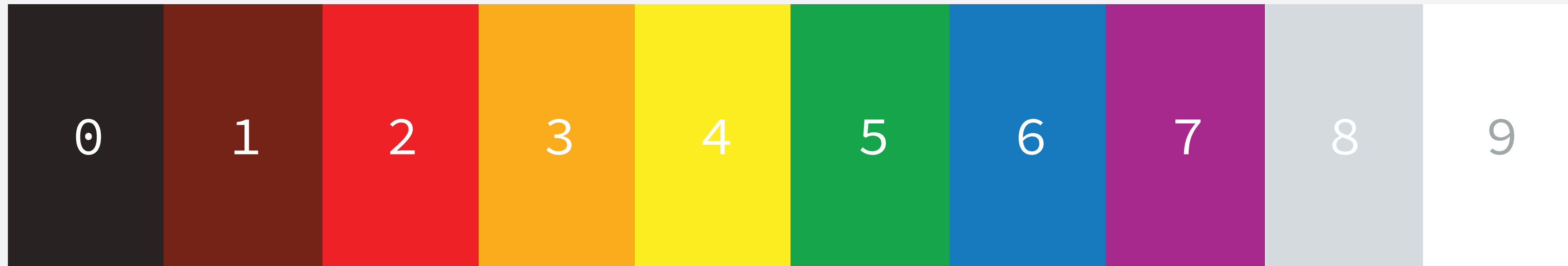
The strength of a resistor can be found using the color bands

Resistors

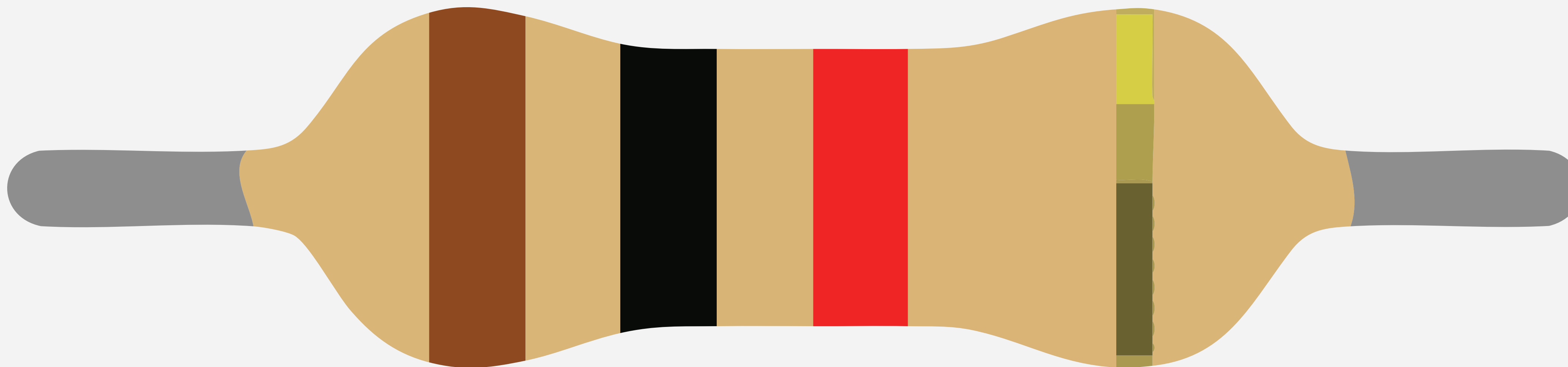


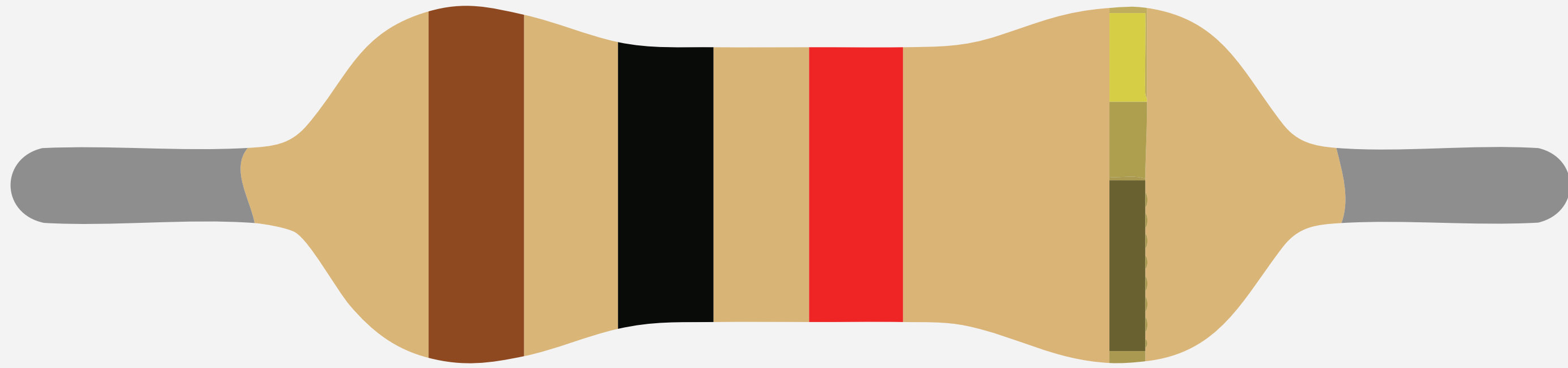
The strength of a resistor can be found using the color bands

Resistors



Black, Brown, ROYGBV, Gray, White

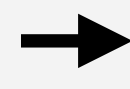




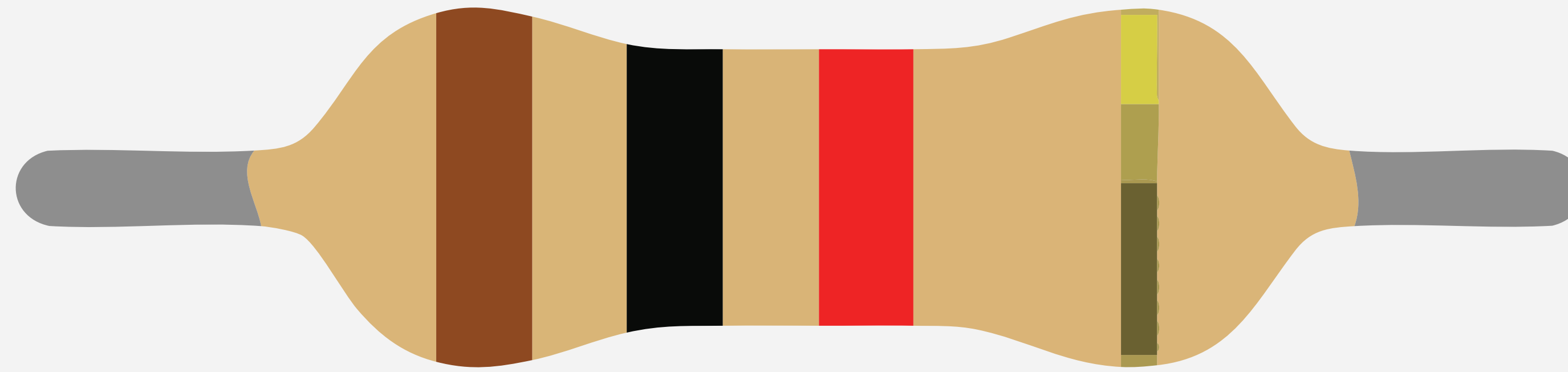
1

= 1

1st DIGIT



- | | |
|---|------------|
| 0 | Dark Grey |
| 1 | Dark Brown |
| 2 | Red |
| 3 | Orange |
| 4 | Yellow |
| 5 | Green |
| 6 | Blue |
| 7 | Purple |
| 8 | Light Grey |
| 9 | White |



1

0

= 10

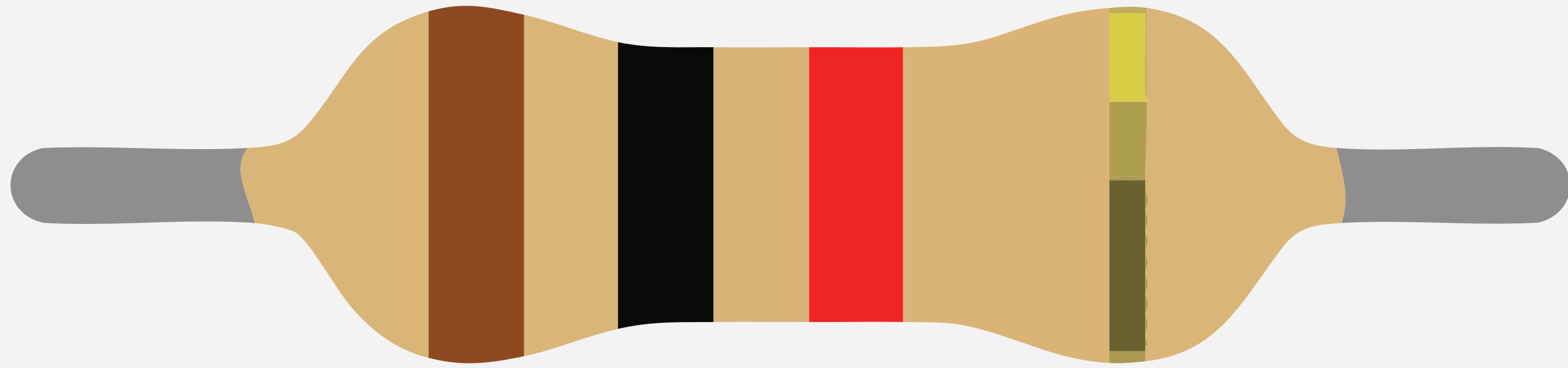
1st DIGIT

2nd DIGIT

0
1
2
3
4
5
6
7
8
9



0
1
2
3
4
5
6
7
8
9



1

0

×

10²

= 10 × 100

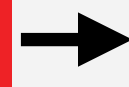
1st DIGIT

2nd DIGIT

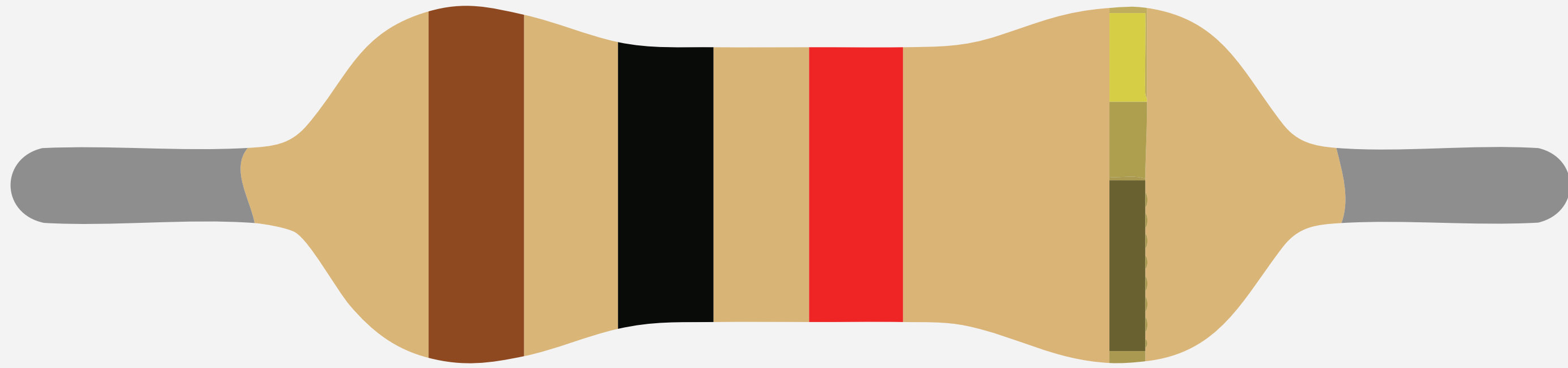
MULTIPLIER

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9



0
1
2
3
4
5
6
7
8
9



1

0

×

10^2

= 1,000

1st DIGIT

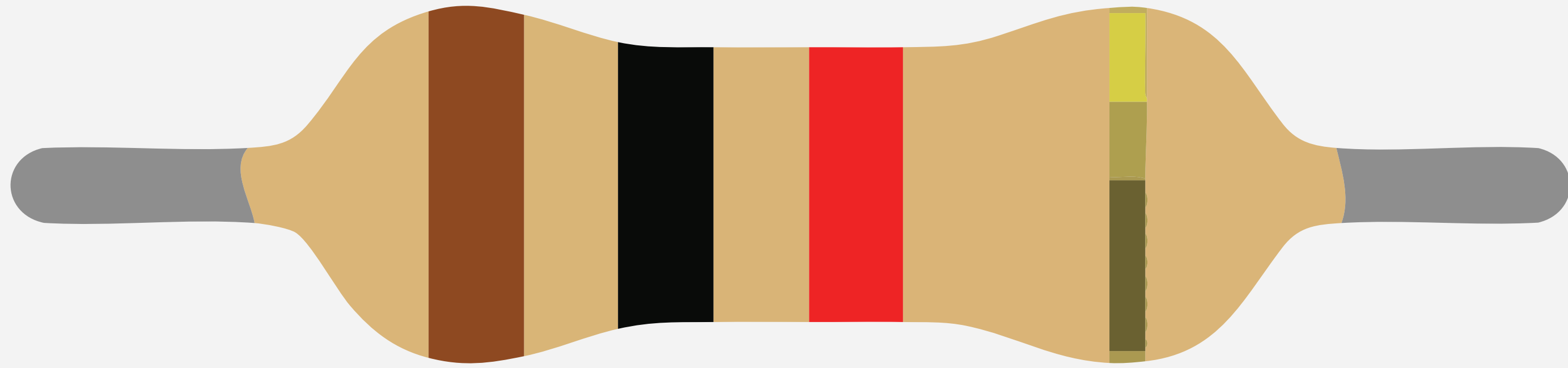
2nd DIGIT

MULTIPLIER

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9



1

0

×

10^2

= 1k

1st DIGIT

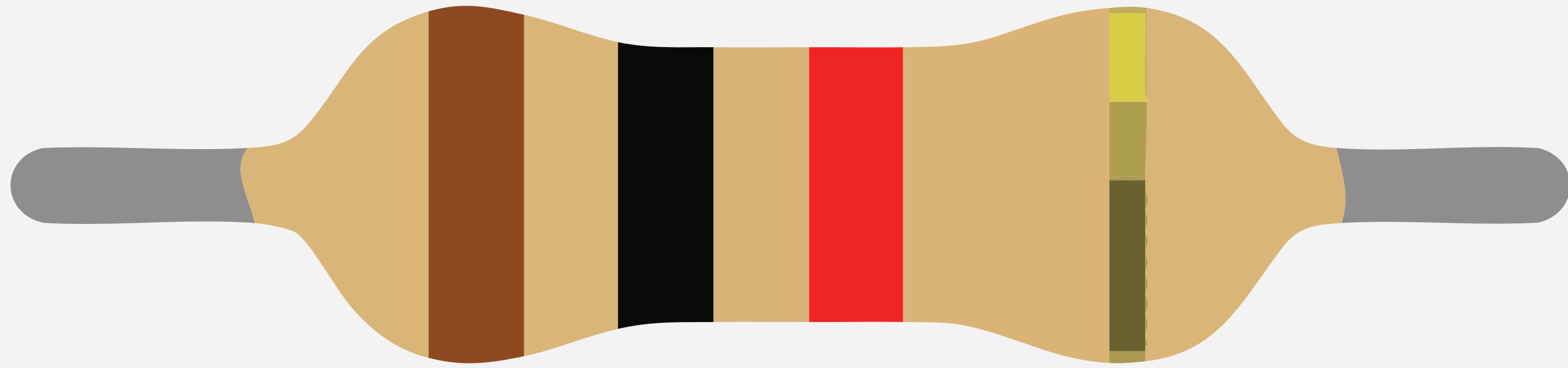
2nd DIGIT

MULTIPLIER

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9



1

0

×

10^2

= 1kΩ

1st DIGIT

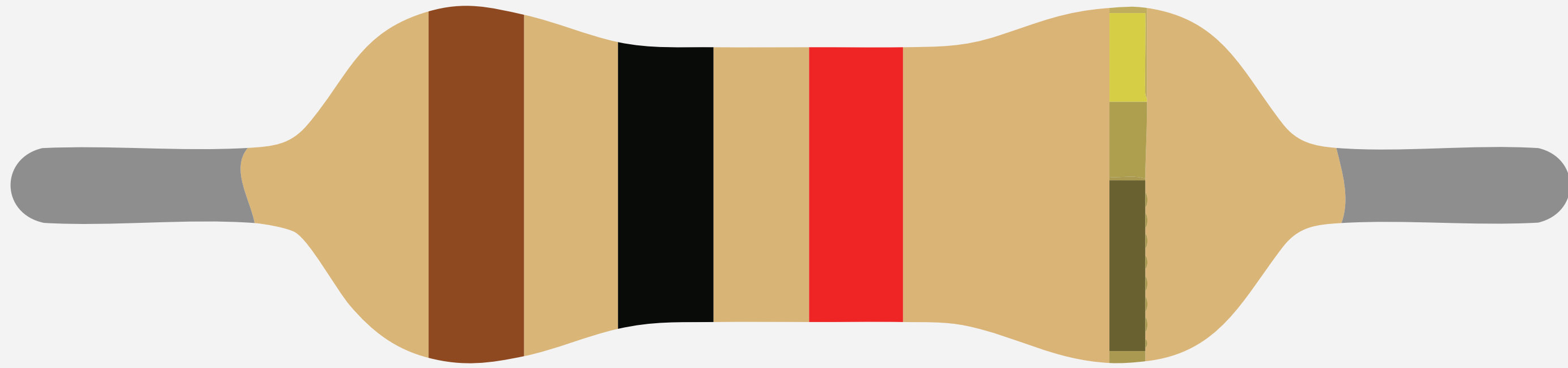
2nd DIGIT

MULTIPLIER

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9



1 0 × 10² ±5 = 1kΩ

1st DIGIT

2nd DIGIT

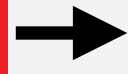
MULTIPLIER

TOLERANCE

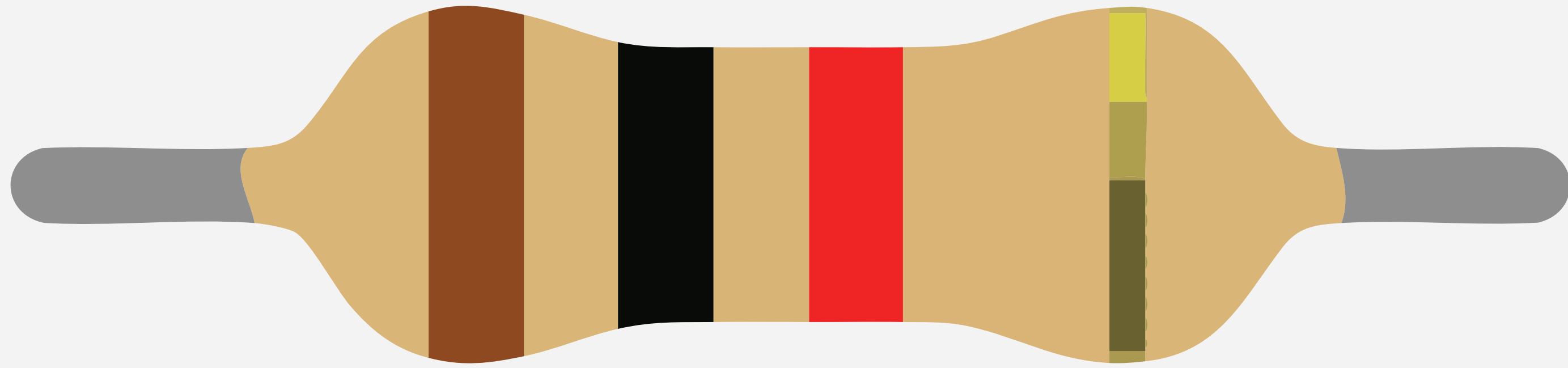
0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9



±1%
±2%
±5% GOLD
±10% SILVER



1

0

×

10^2

±5

= 1kΩ±5%

1st DIGIT

2nd DIGIT

MULTIPLIER

TOLERANCE

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9

0
1
2
3
4
5
6
7
8
9

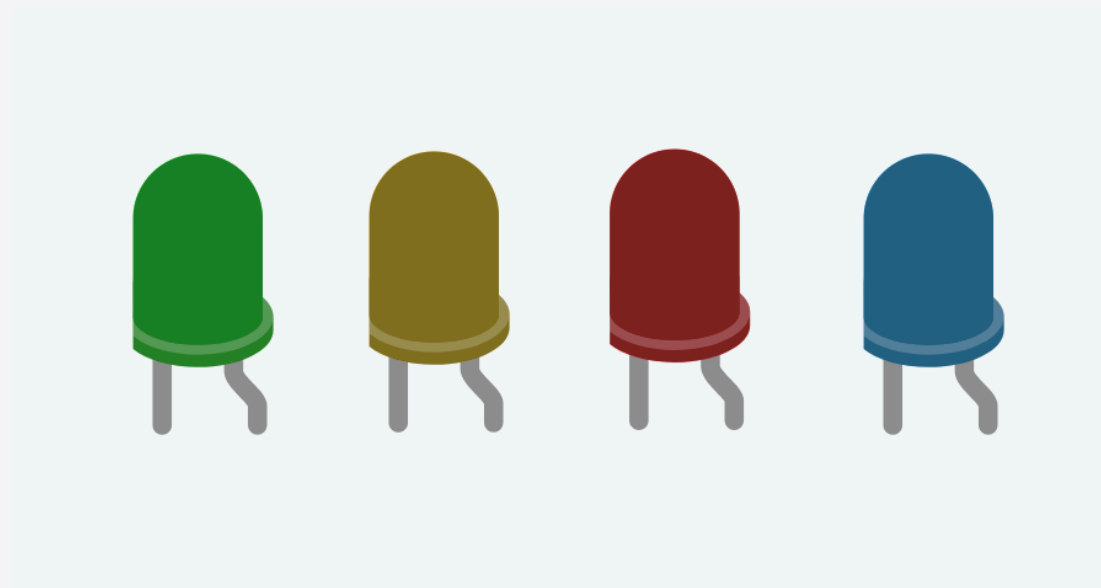
±1%
±2%
±5% GOLD
±10% SILVER

We will use resistors to
create the right amount of
current for LEDs and to
create and calibrate sensors.



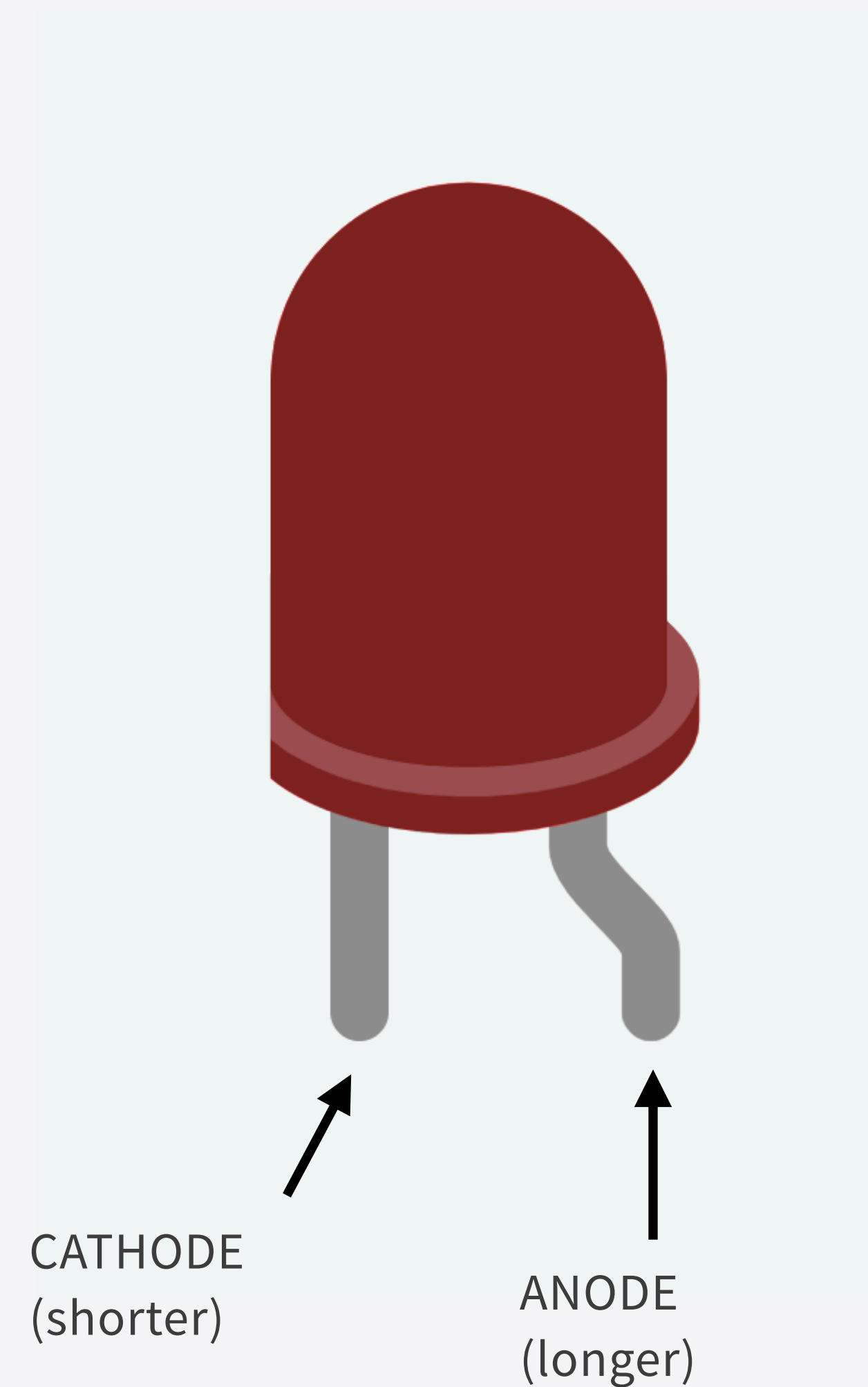
LED

LED

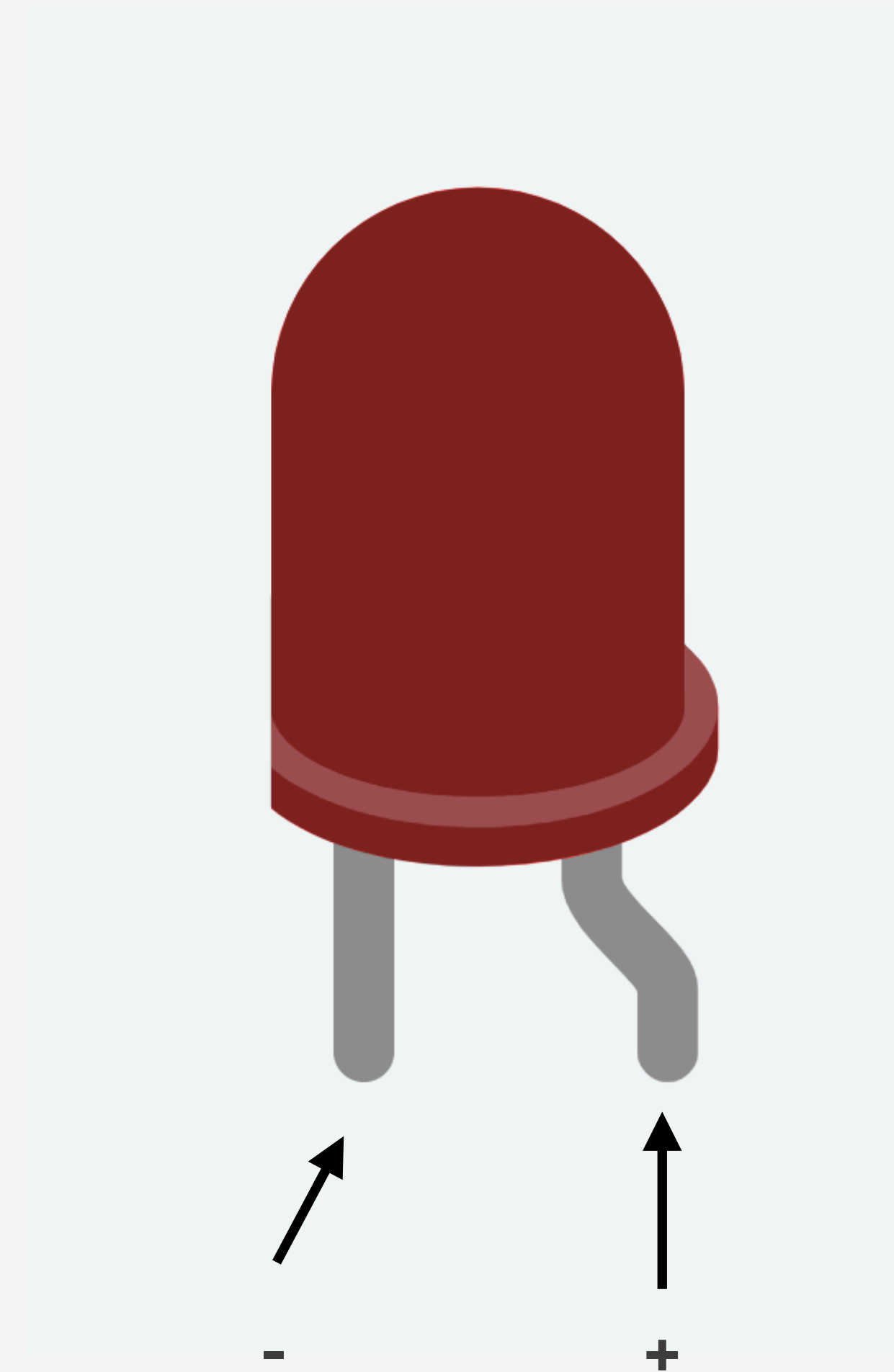


A light-emitting diode (LED) is a semiconductor light source that emits light when current flows through it.

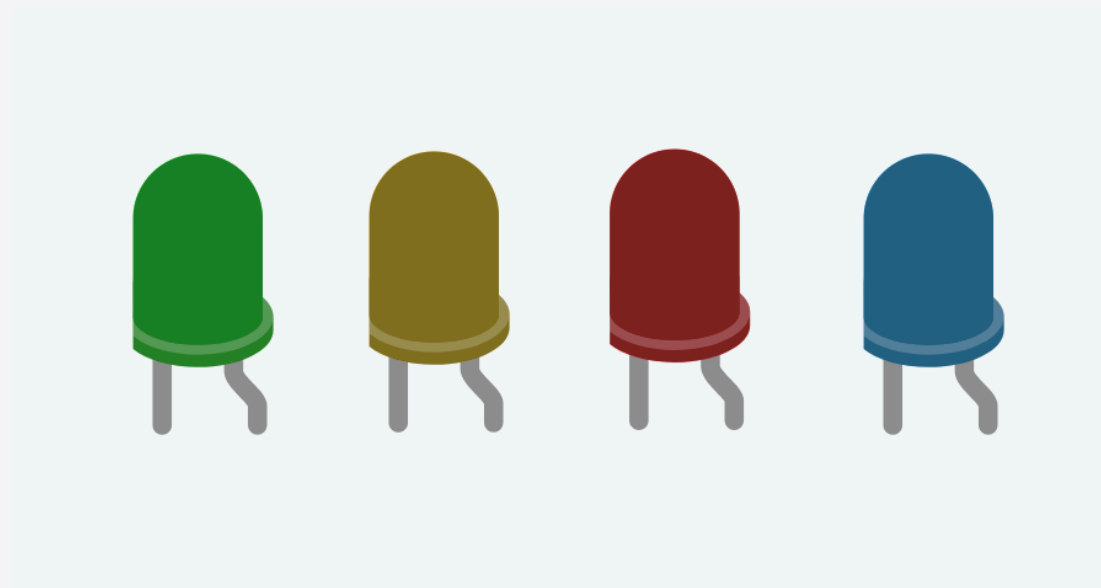
An LED has two legs, a longer one called an anode and a shorter one called a cathode



The cathode
must be
connected to
ground, and the
anode
connected to
power

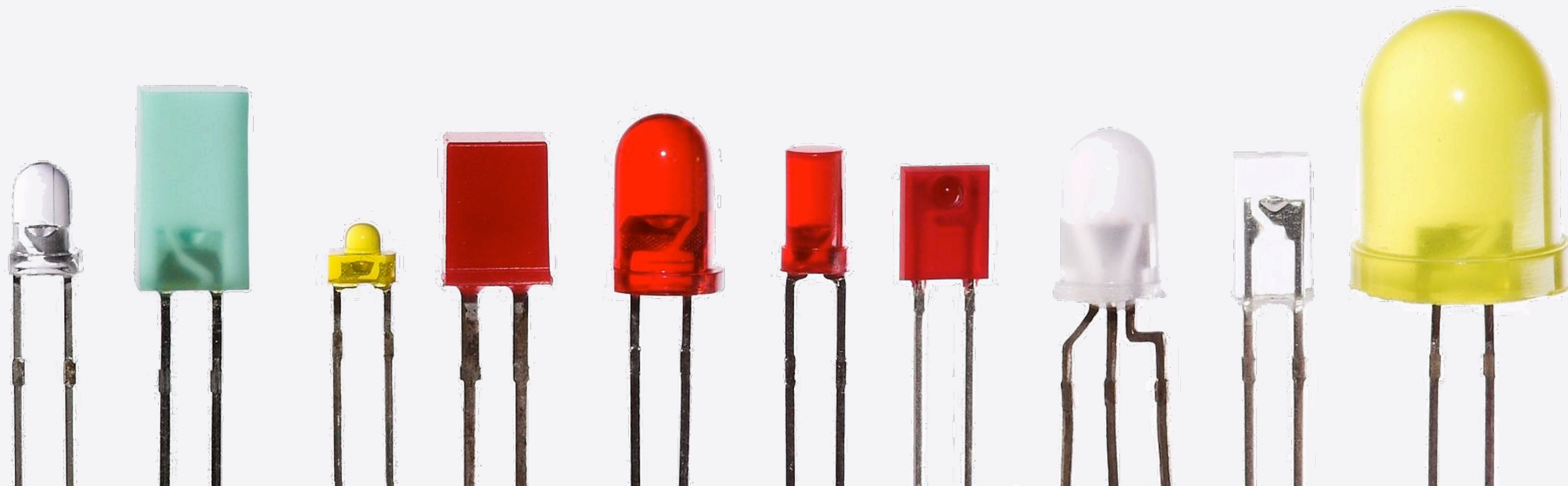


LED



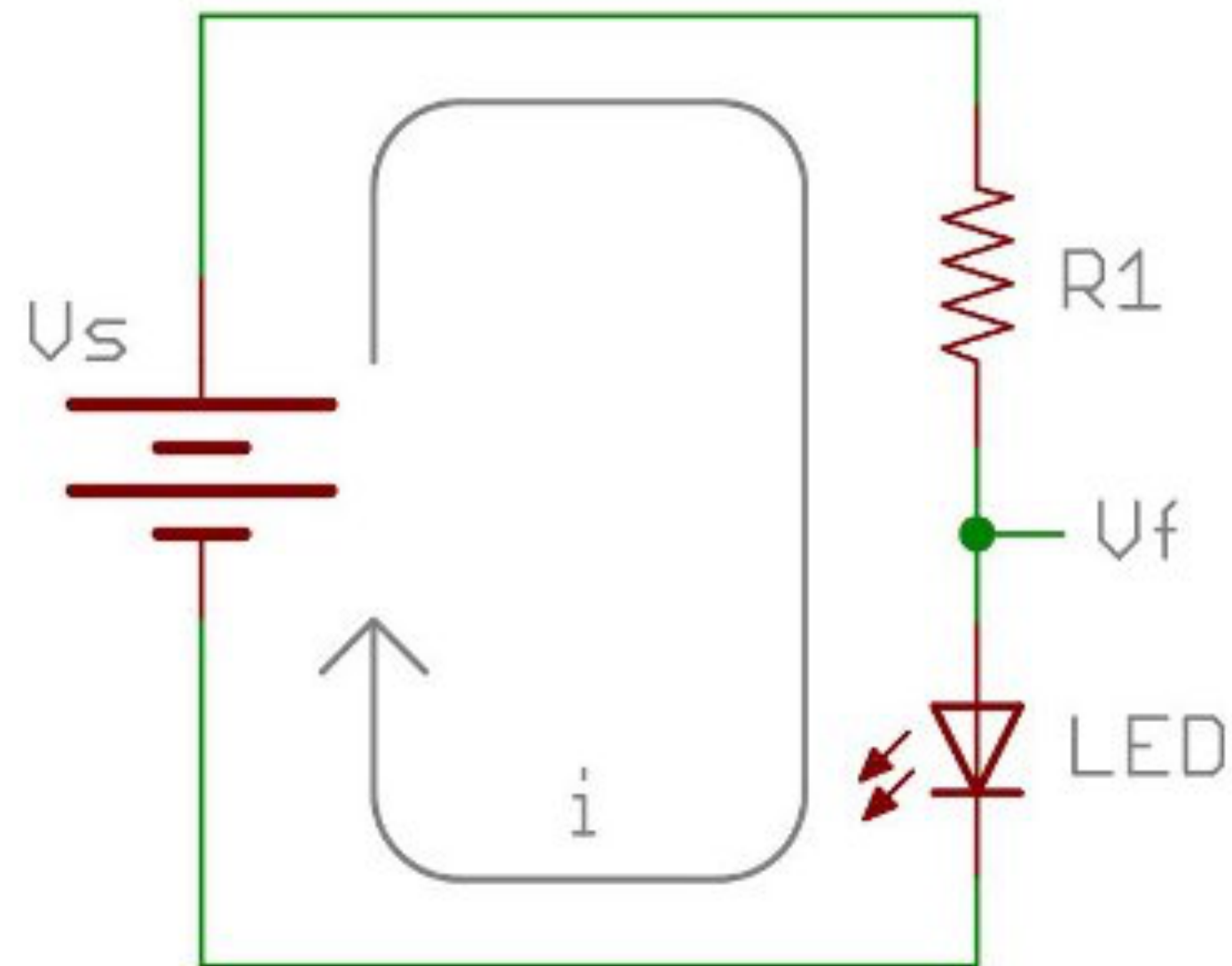
Every LED has a rating for how much current it can handle, and how much voltage is required for it to work

LEDs come in many shapes and sizes. The color is often indicated by the color of the plastic, but not always.



Using LEDs

$$R = \frac{V_s - V_f}{i}$$



i = LED forward current in Amps
(found in the LED datasheet)

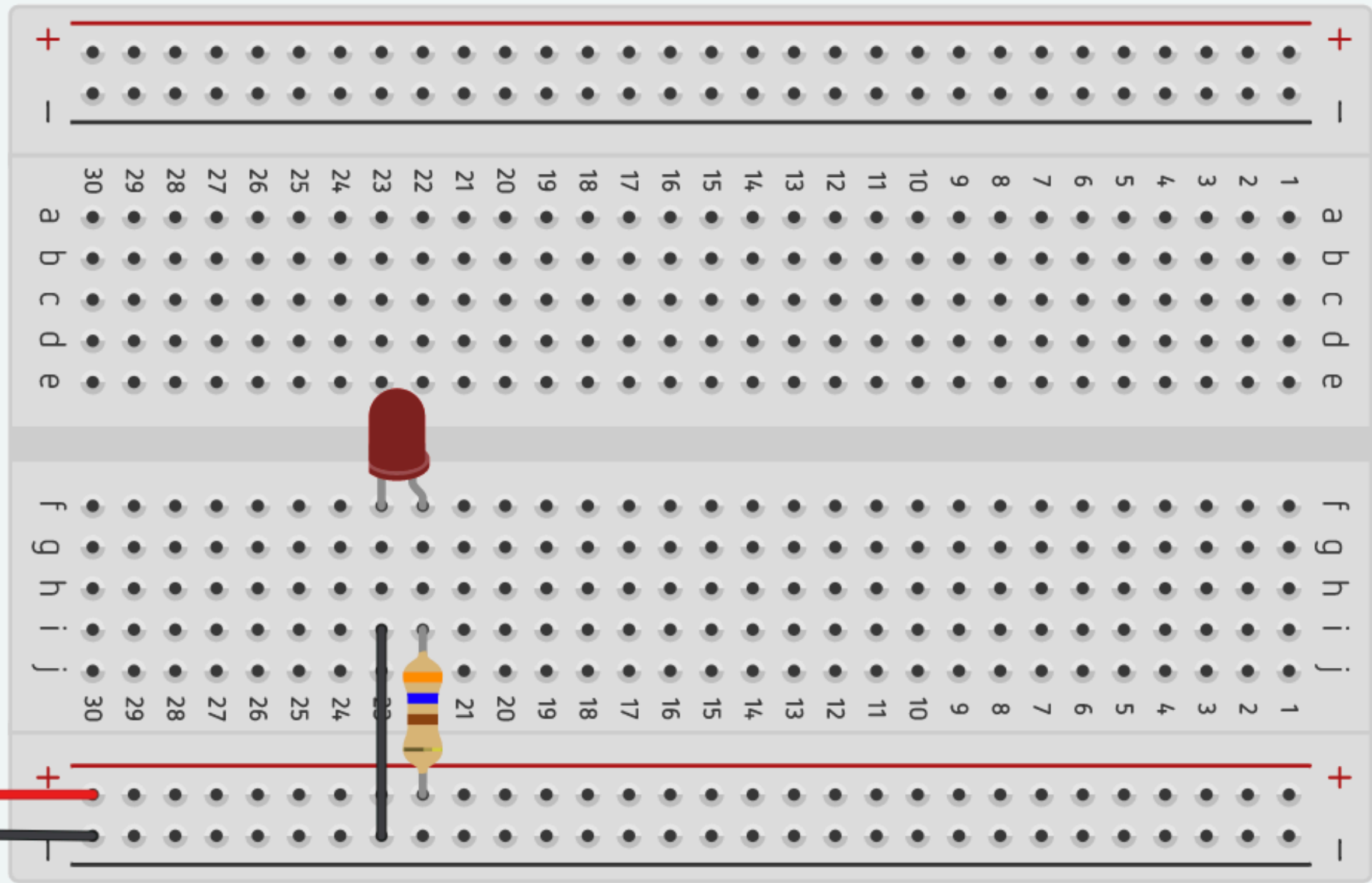
V_f = LED forward voltage drop in Volts
(found in the LED datasheet)

V_s = supply voltage

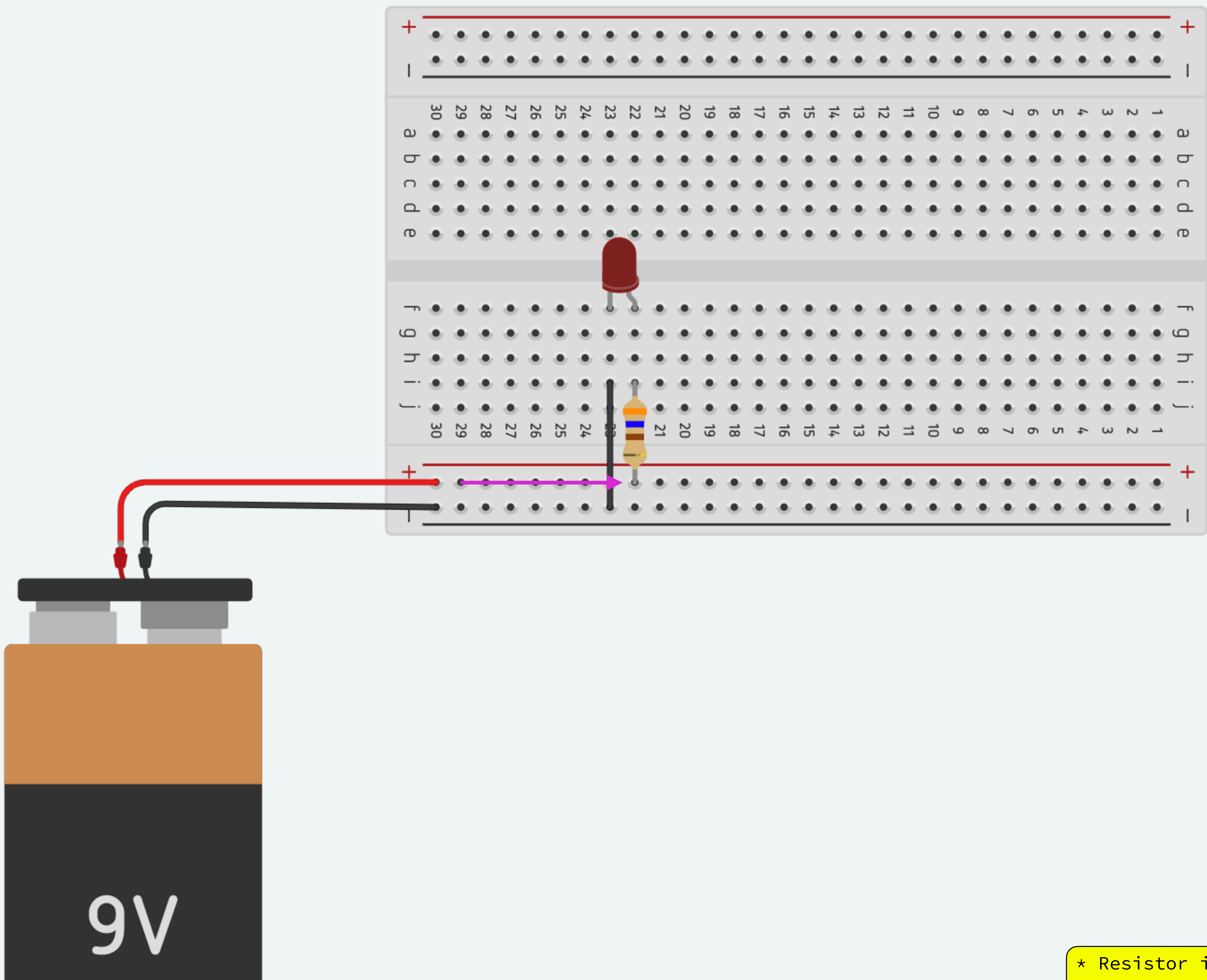
$$R = \frac{V_s - V_f}{i}$$



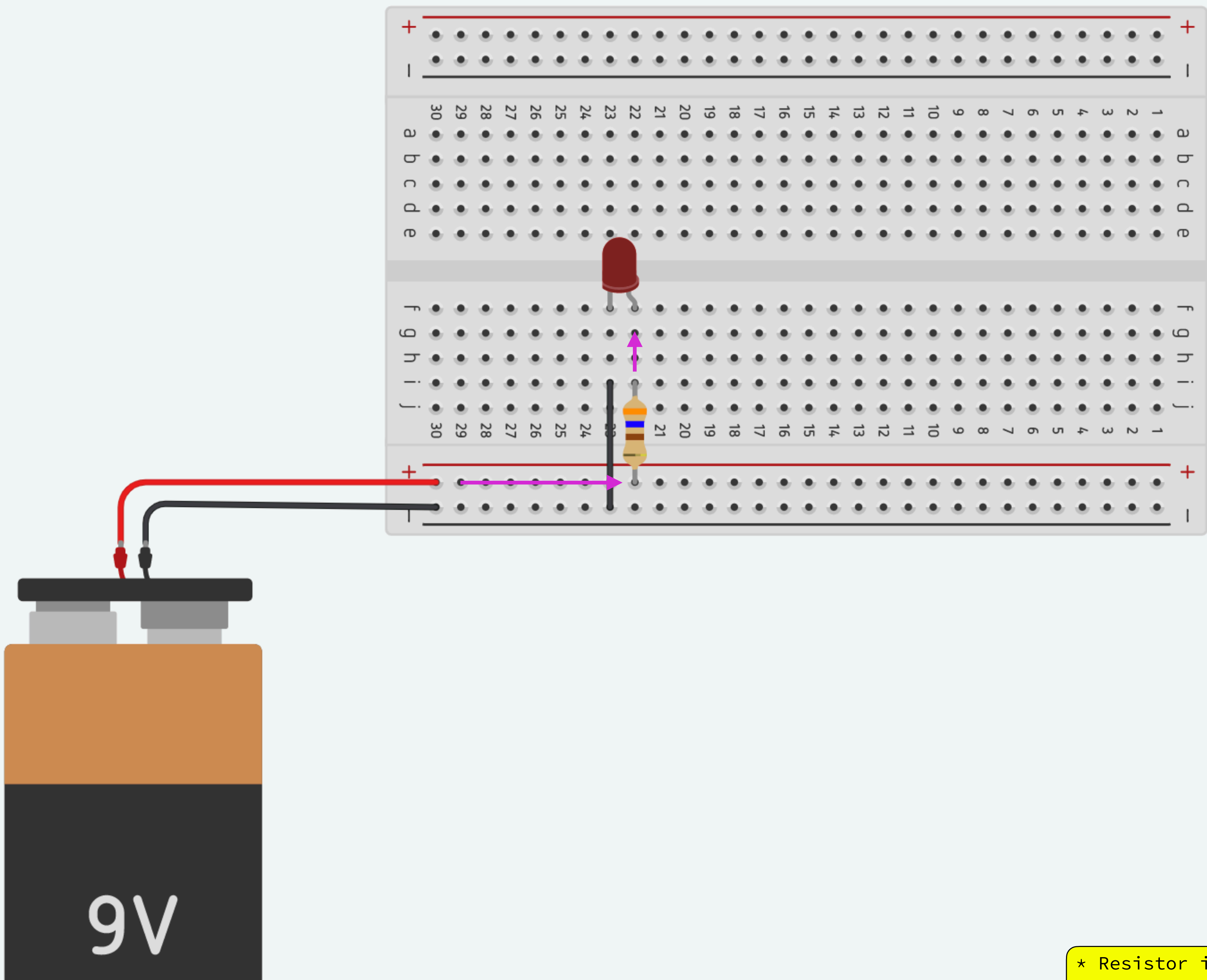
<https://ledcalculator.net/>



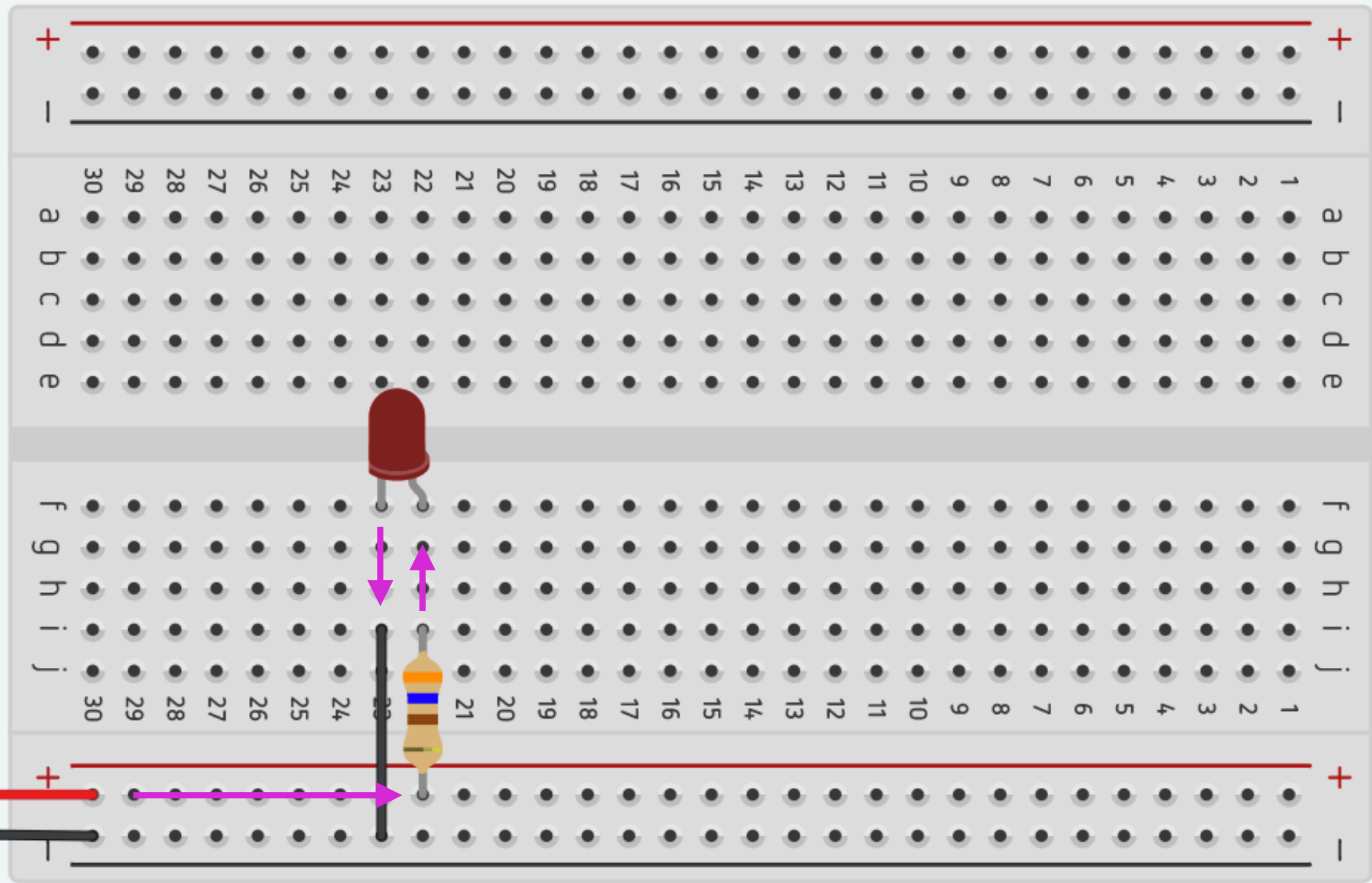
* Resistor is 360Ω (Orange Blue Brown)



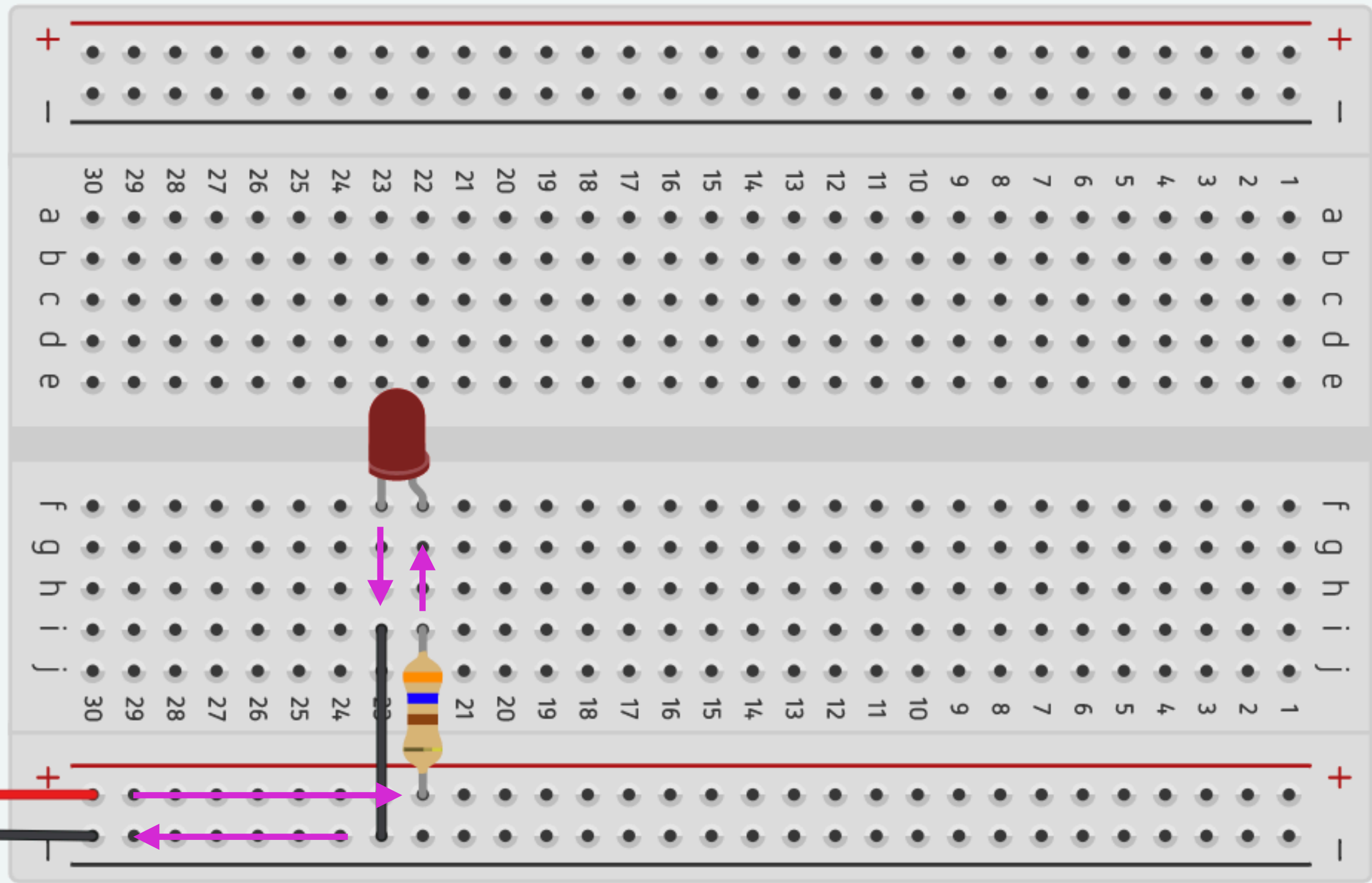
* Resistor is 360Ω (Orange Blue Brown)



* Resistor is 360Ω (Orange Blue Brown)

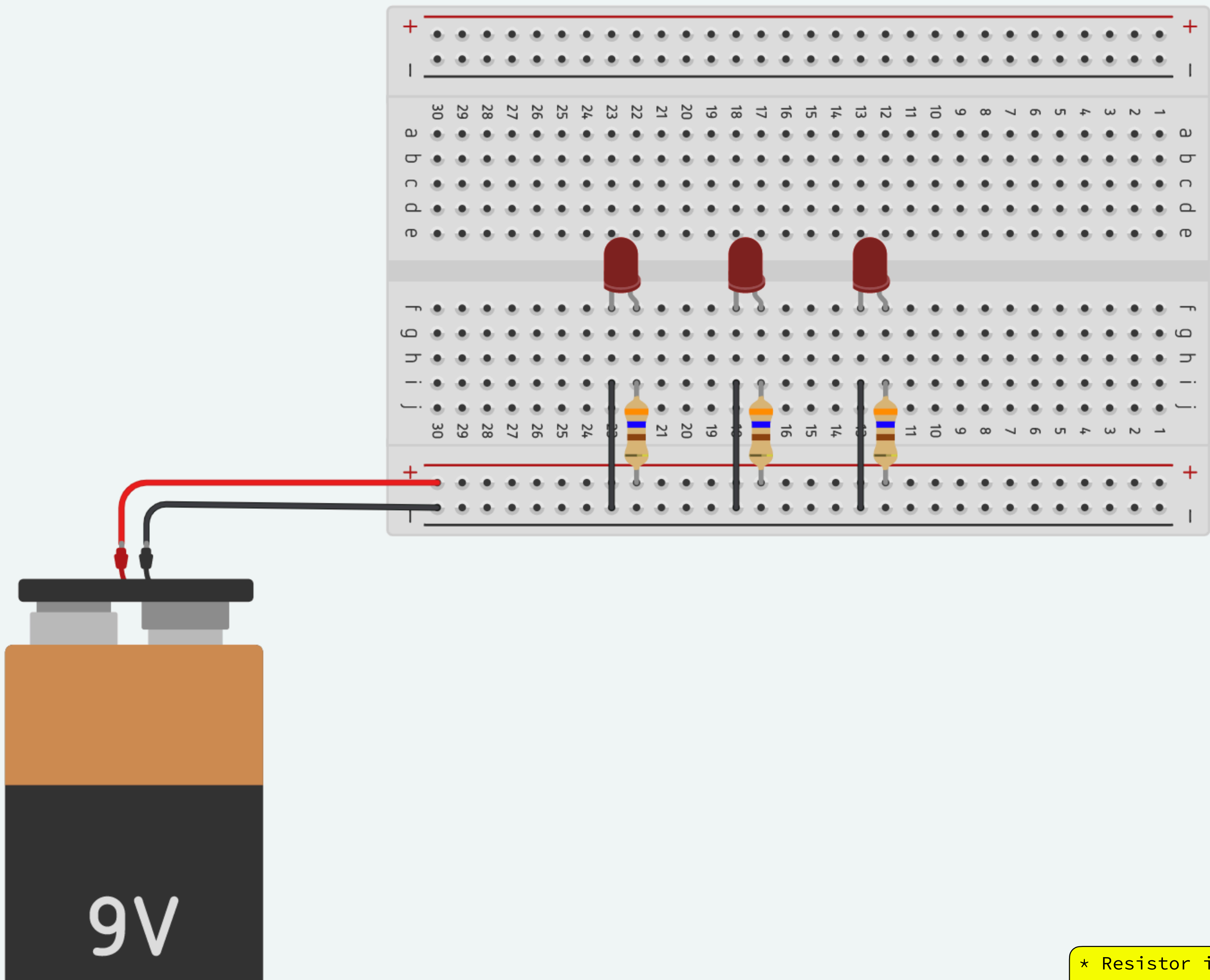


* Resistor is 360Ω (Orange Blue Brown)



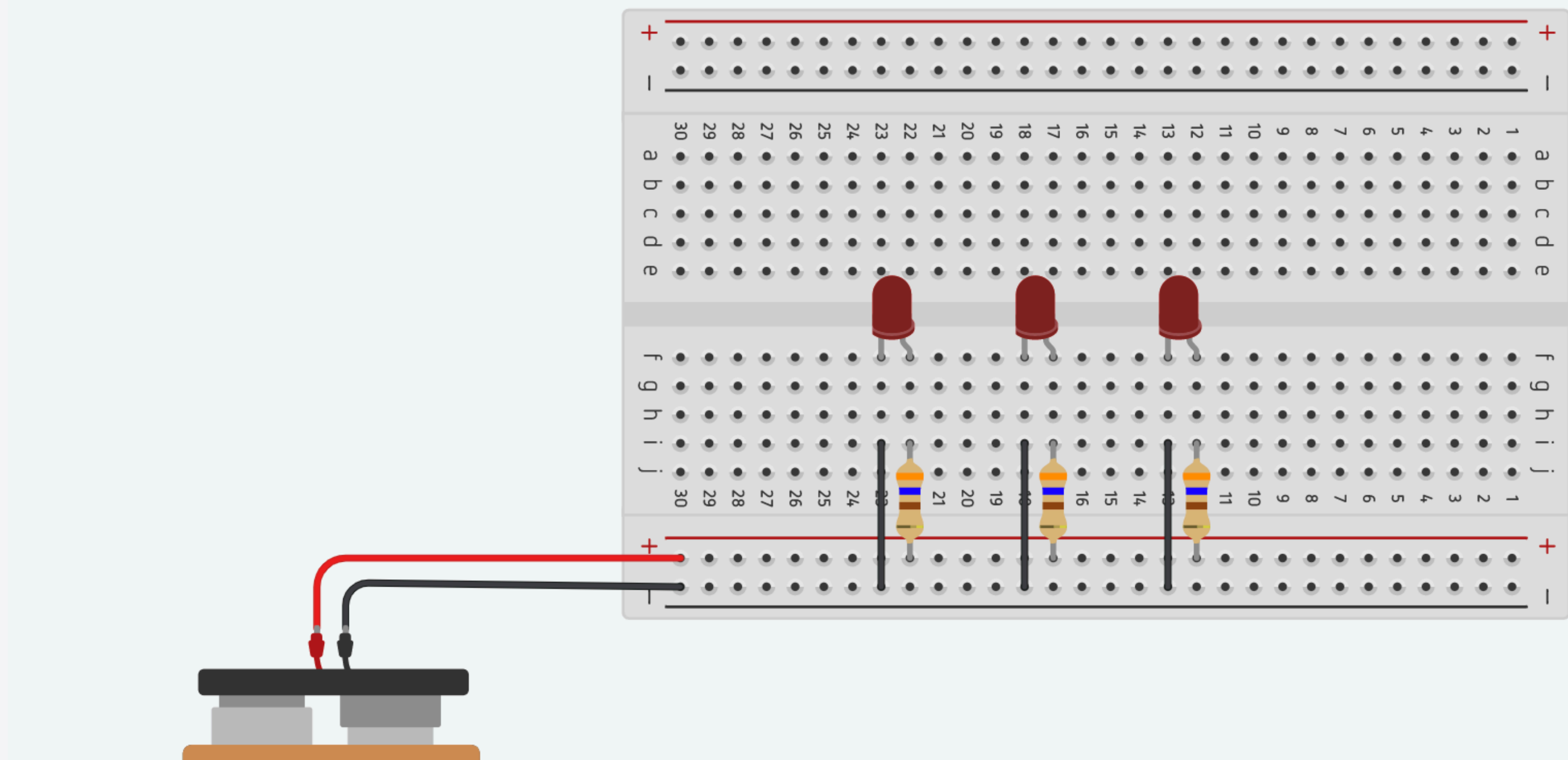
* Resistor is 360Ω (Orange Blue Brown)

Try adding more LEDs

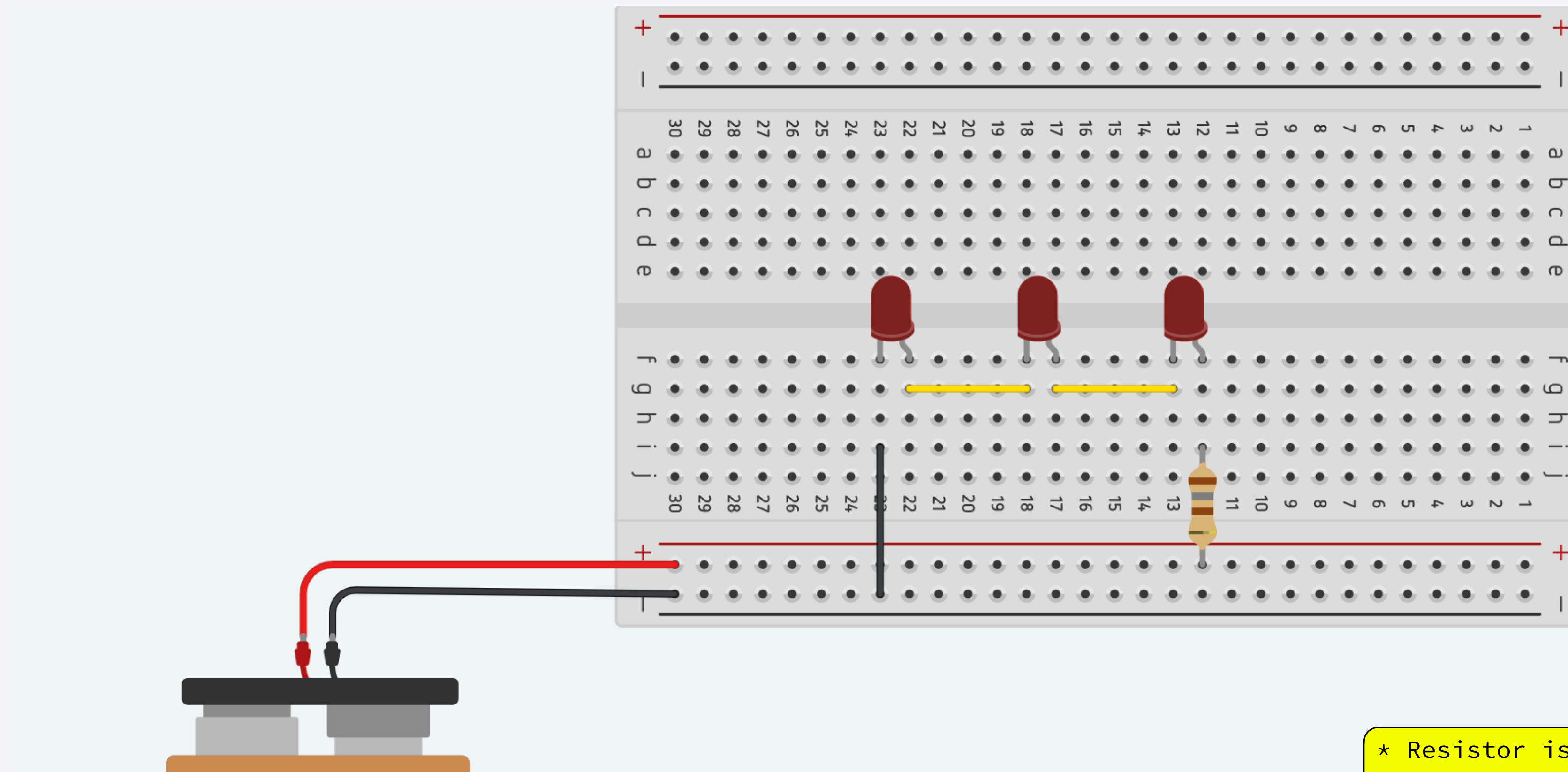


* Resistor is 360Ω (Orange Blue Brown)

This layout is called parallel because it is as-if there are 3 separate circuits, one for each LED

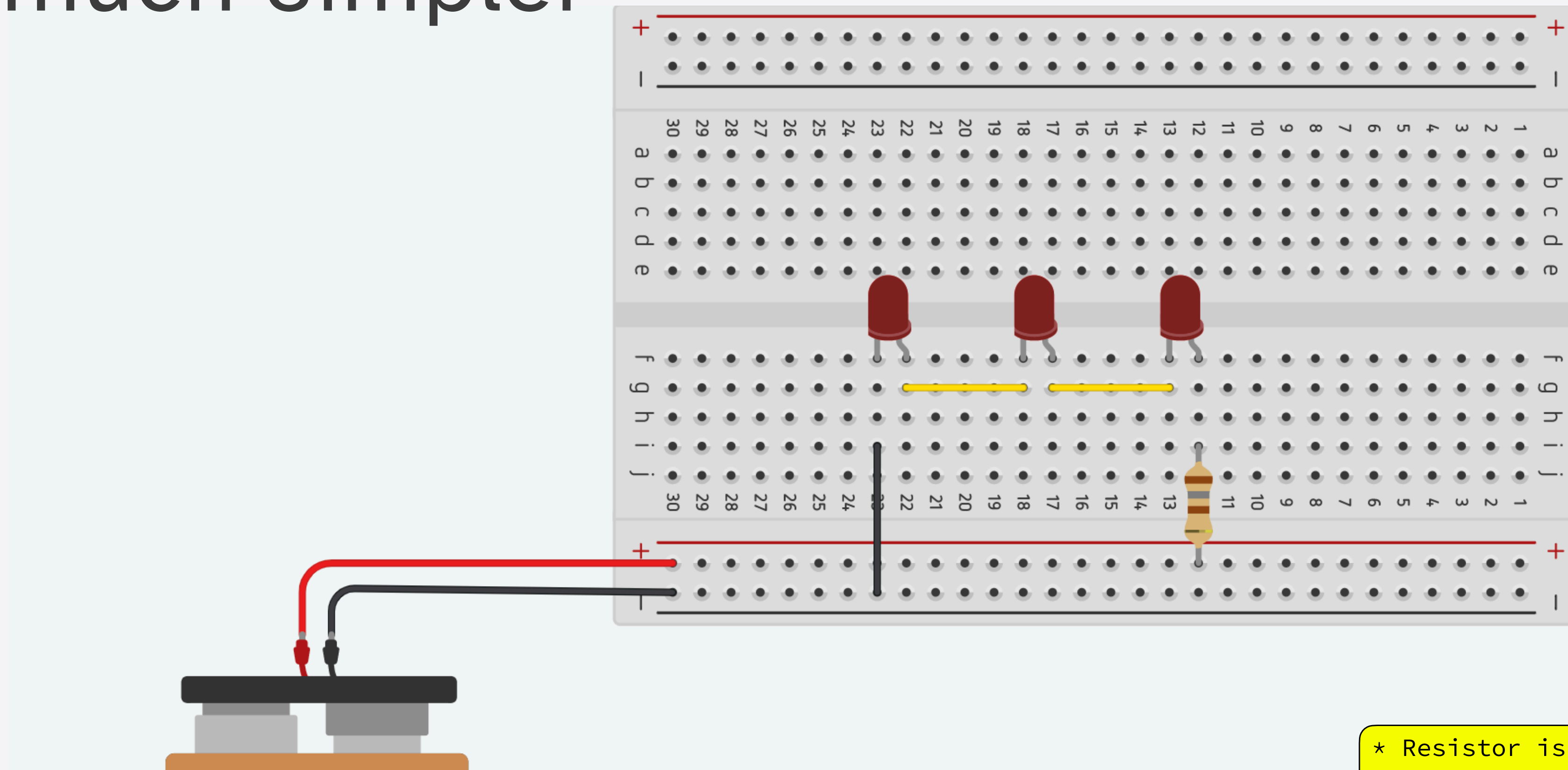


You can also connect them in series, with each LED connected to the next, and a smaller resistor.



* Resistor is 180Ω (Brown Gray Brown)

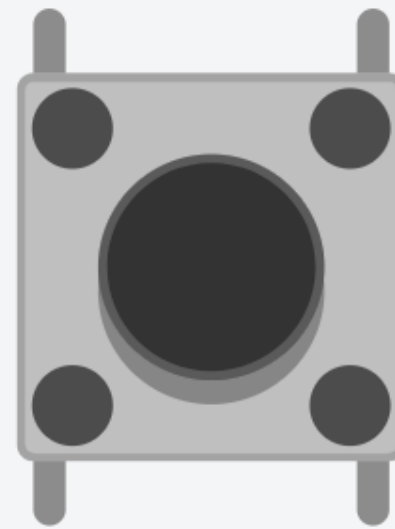
Connecting them in series is more likely to fail because if one component breaks, they all stop working, however it uses less components and the wiring is much simpler



* Resistor is 180Ω (Brown Gray Brown)

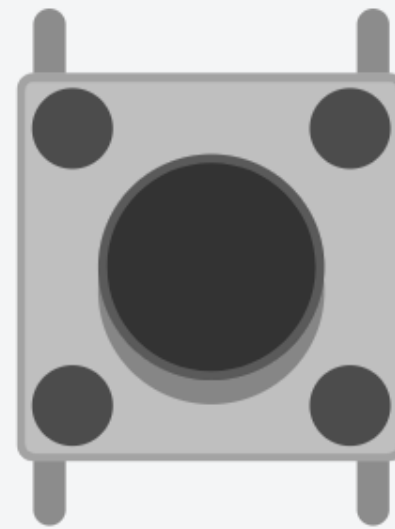
Switches and Buttons

Buttons



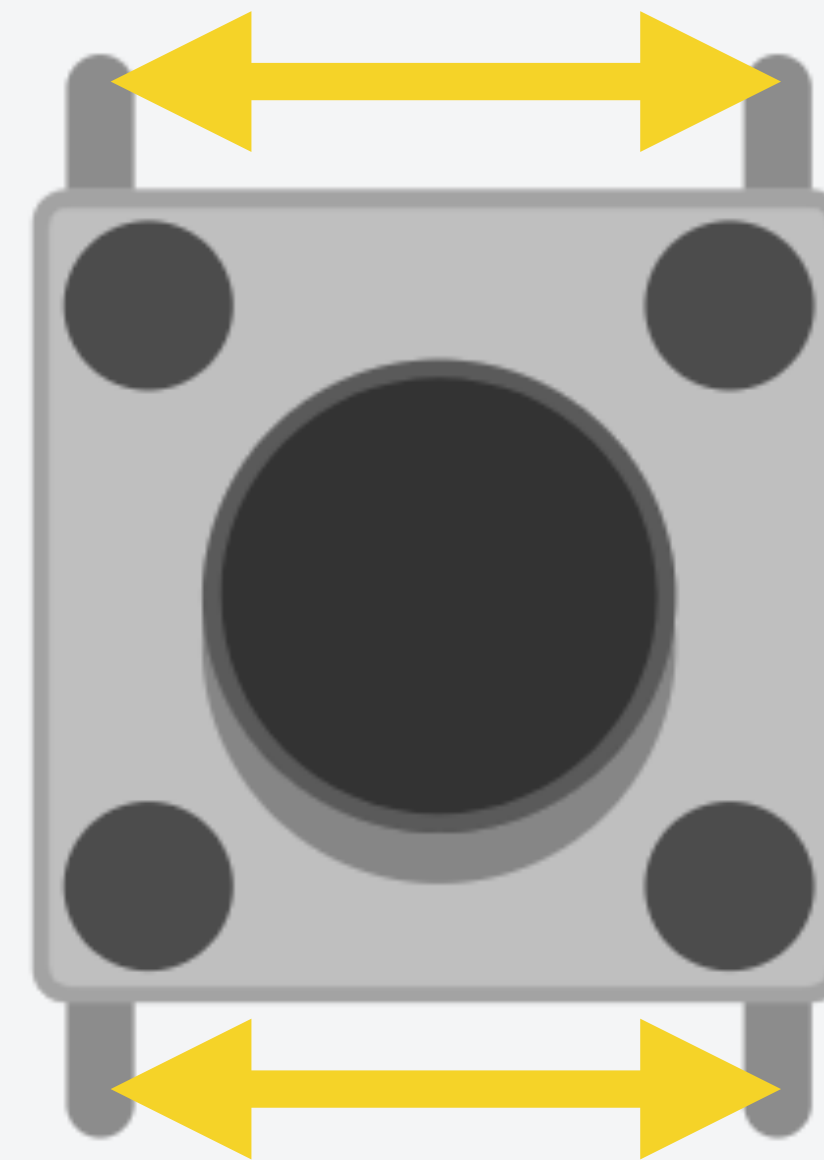
A button is a simple switch mechanism that completes or breaks a circuit when pushed.

Buttons



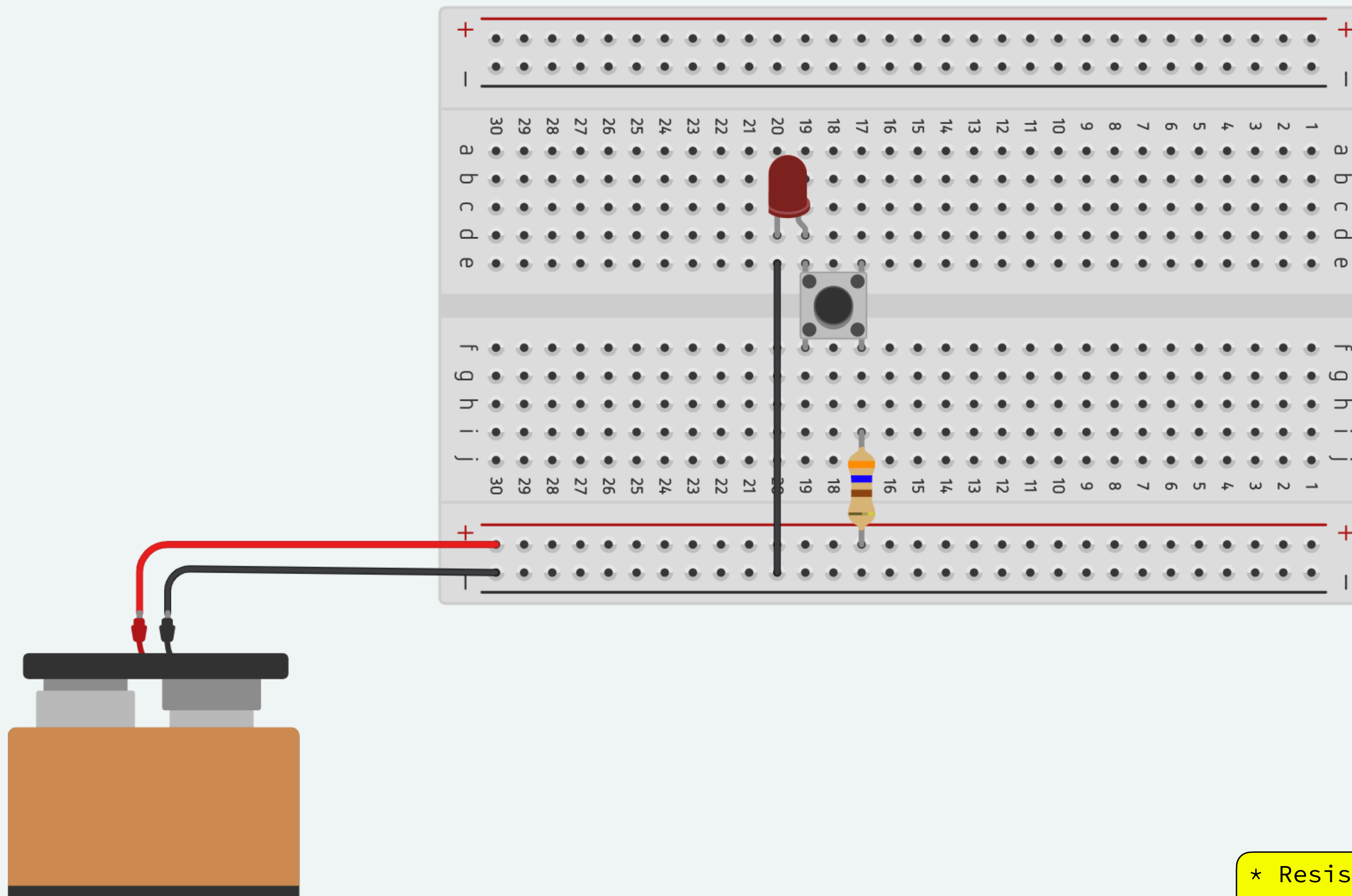
Buttons typically come in two types normally closed (NC) and normally open (NO). NO buttons complete the circuit when pressed, NC buttons break the circuit when pressed.

When pressed, a NO button connects the legs from one side to those on the other side. Most of the buttons we will use are NO



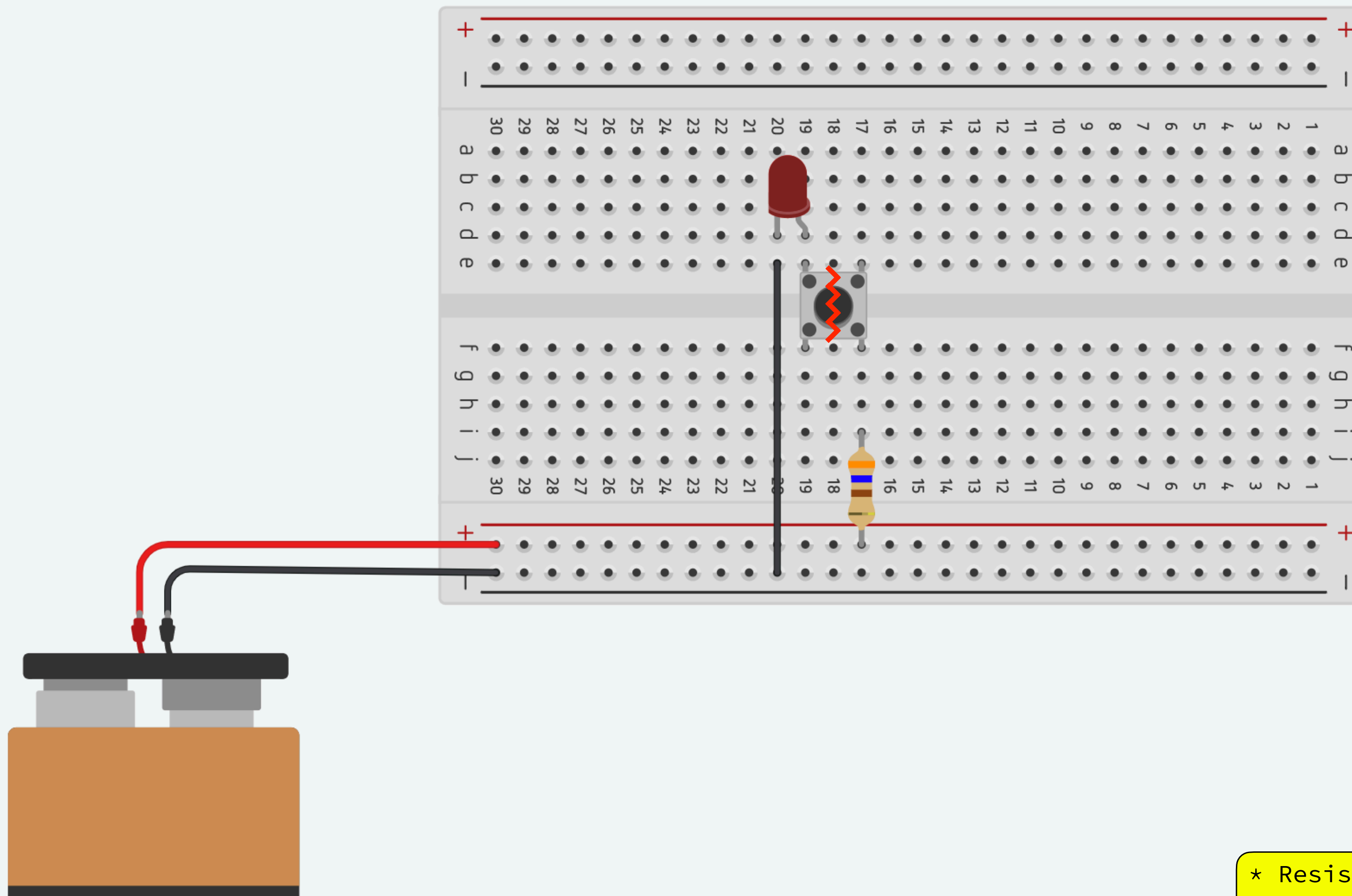
Using Switches (Buttons) with LEDS

Our first LED circuit with a button. The LED will light when the button is pressed.



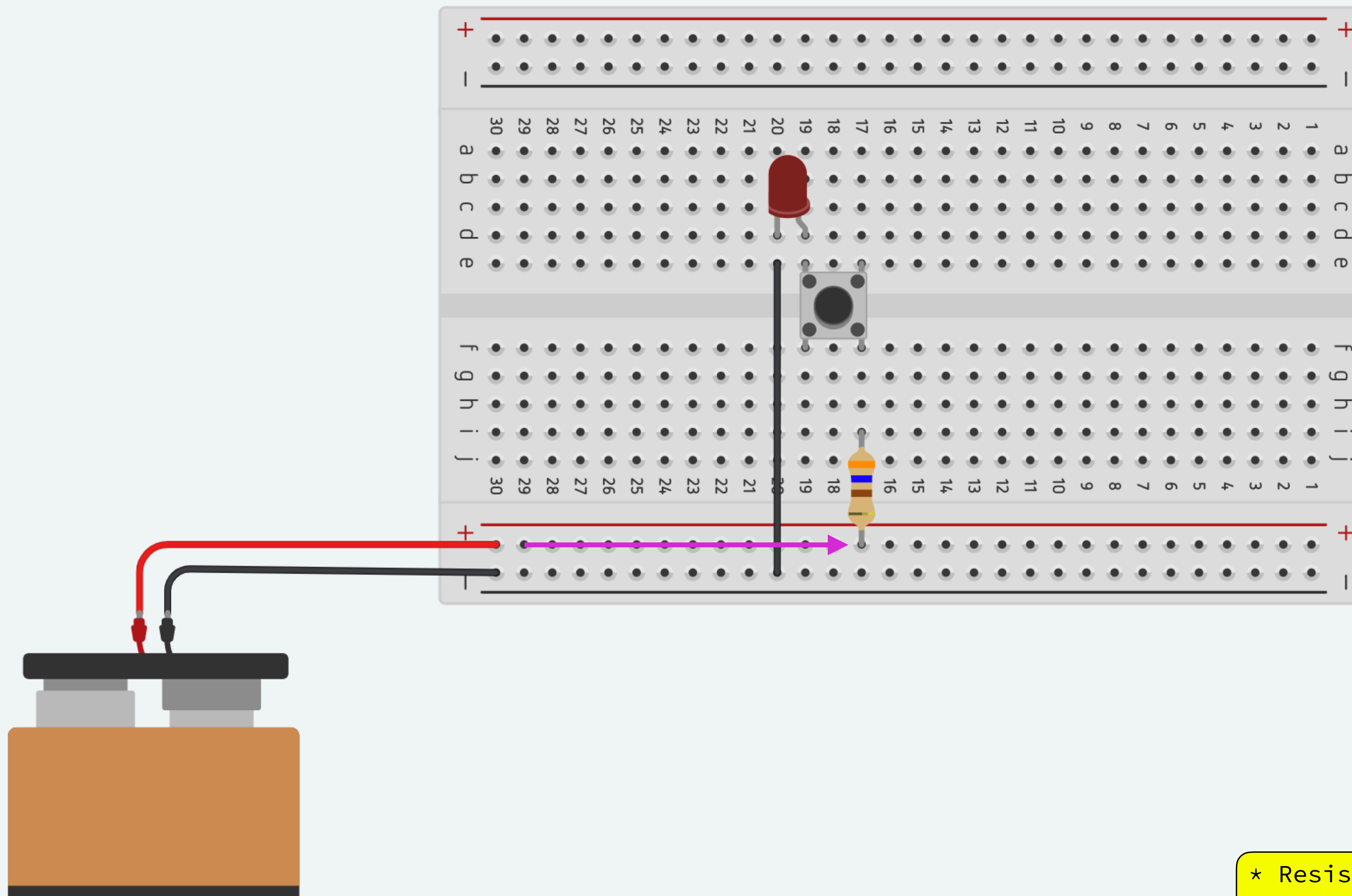
* Resistor is 360Ω (Orange Blue Brown)

Our first LED circuit with a button.
The LED will light when the button
is pressed.



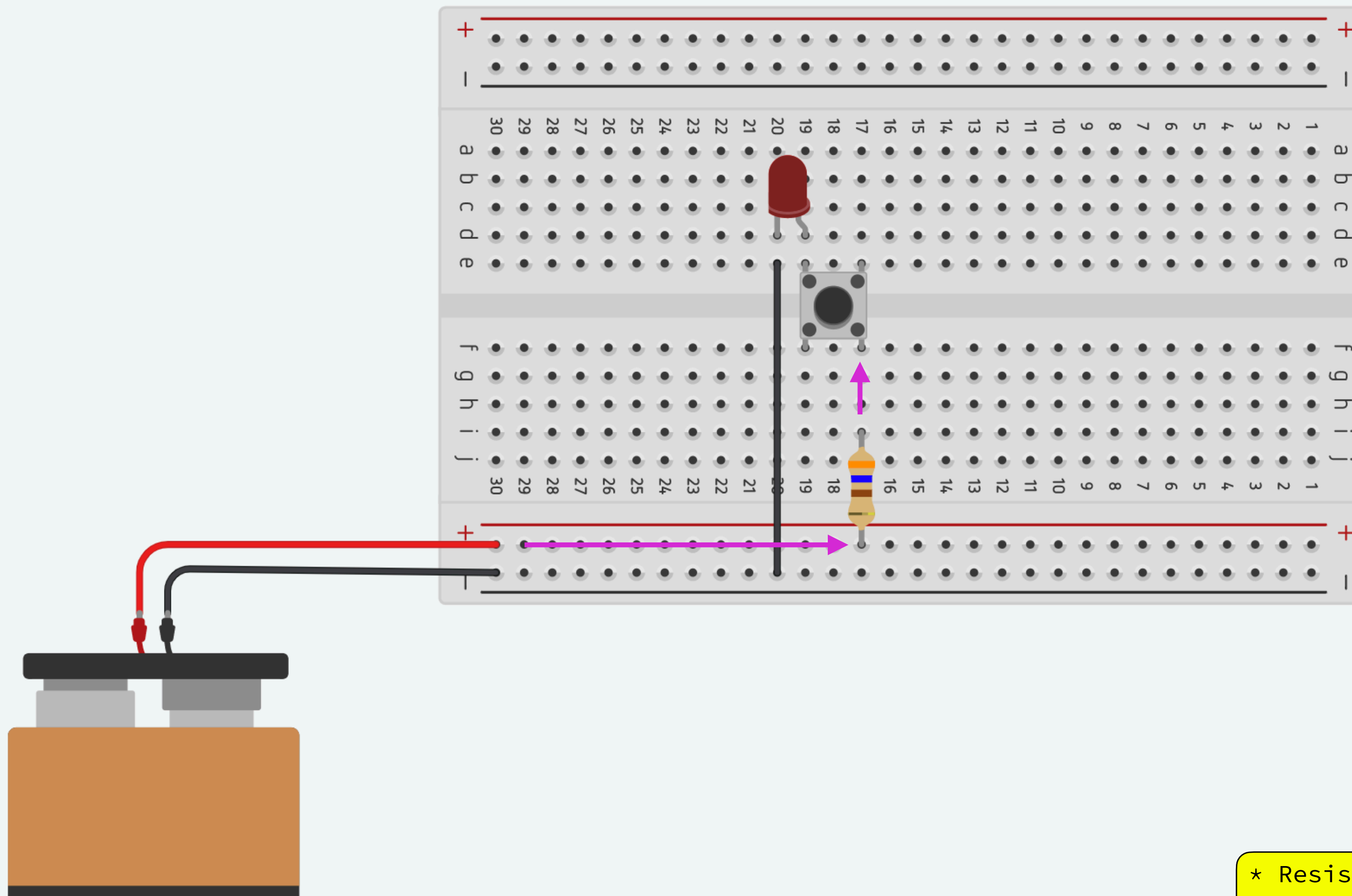
* Resistor is 360Ω (Orange Blue Brown)

Our first LED circuit with a button. The LED will light when the button is pressed.



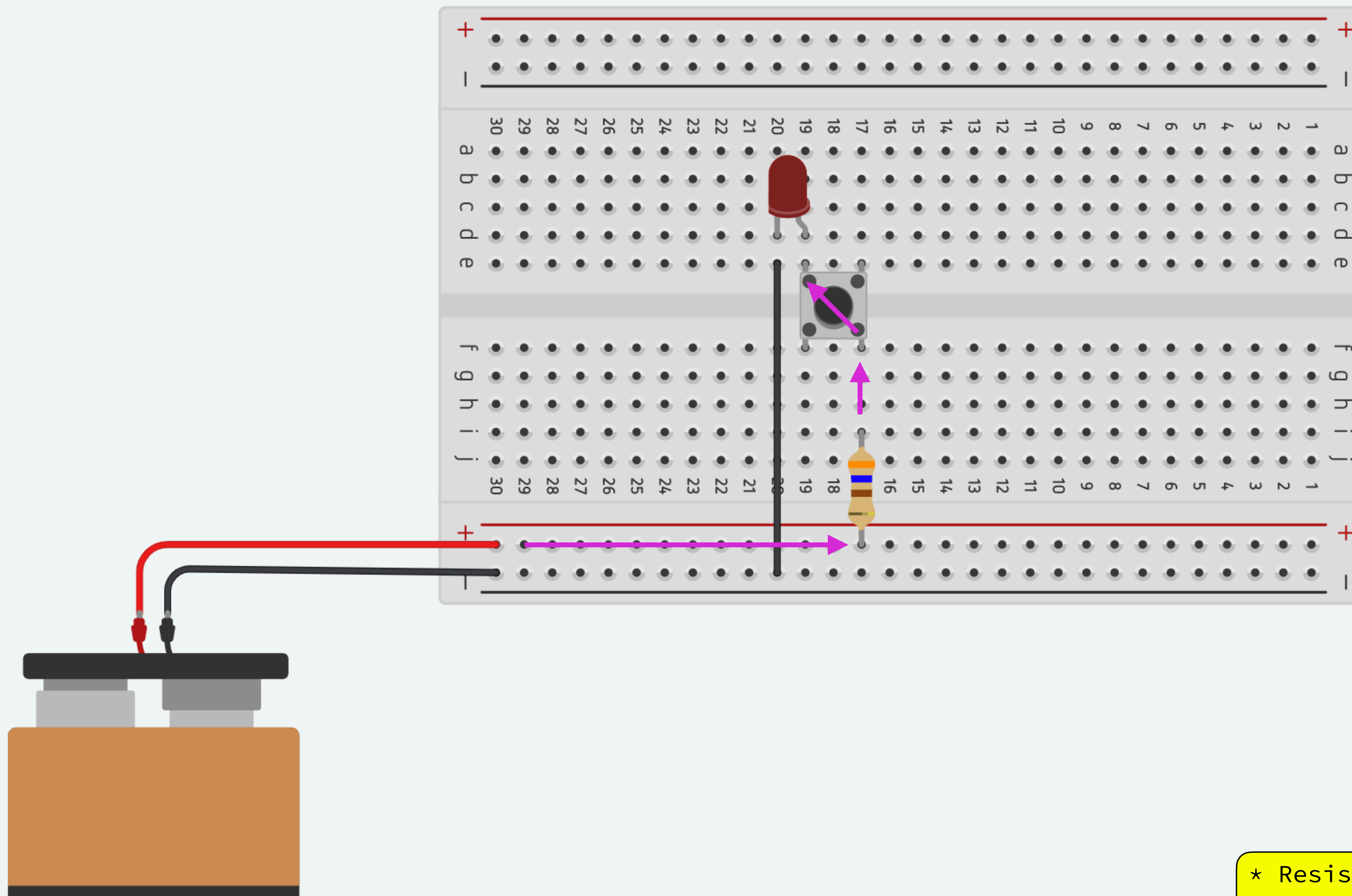
* Resistor is 360Ω (Orange Blue Brown)

Our first LED circuit with a button. The LED will light when the button is pressed.



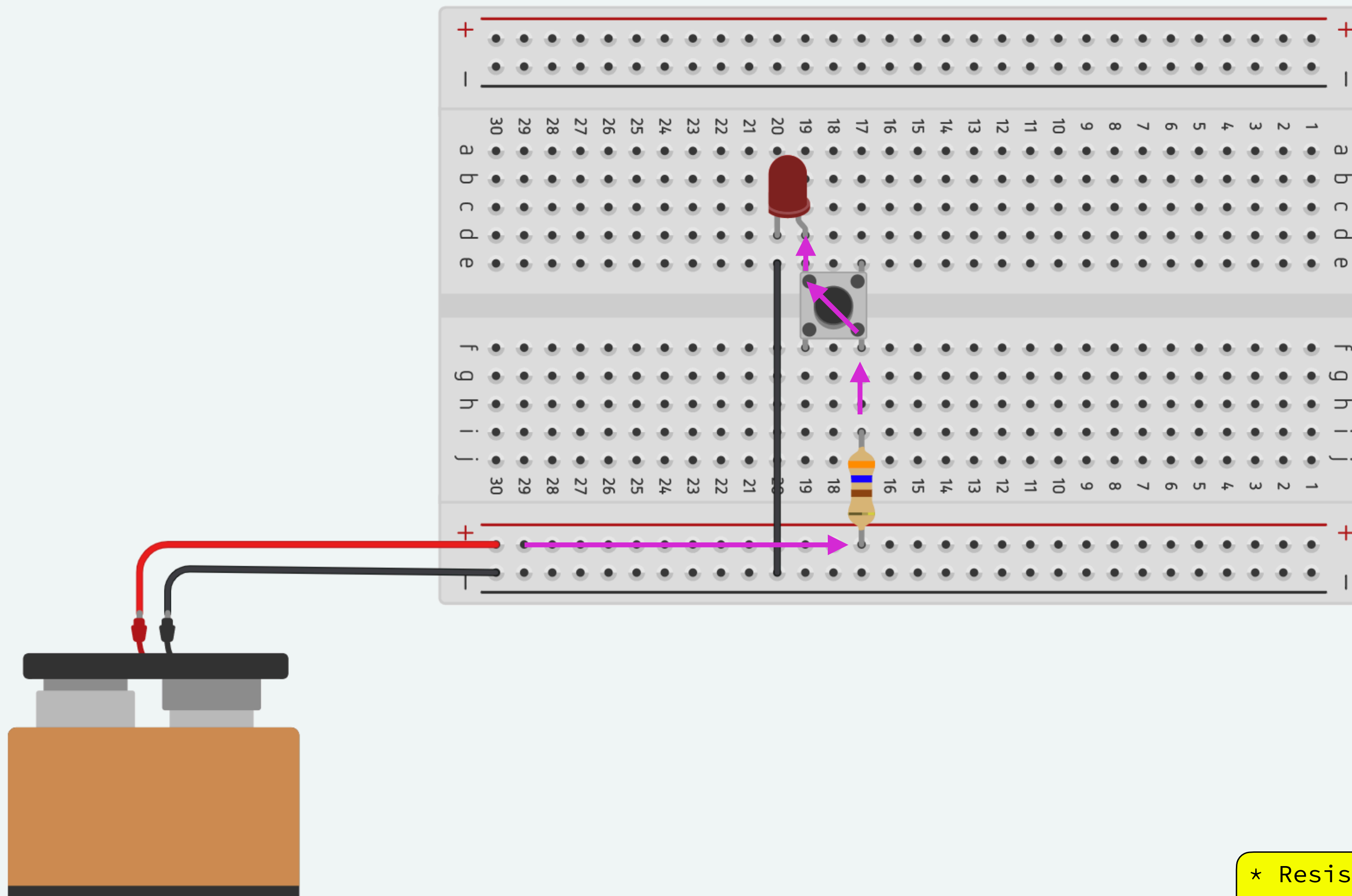
* Resistor is 360Ω (Orange Blue Brown)

Our first LED circuit with a button. The LED will light when the button is pressed.



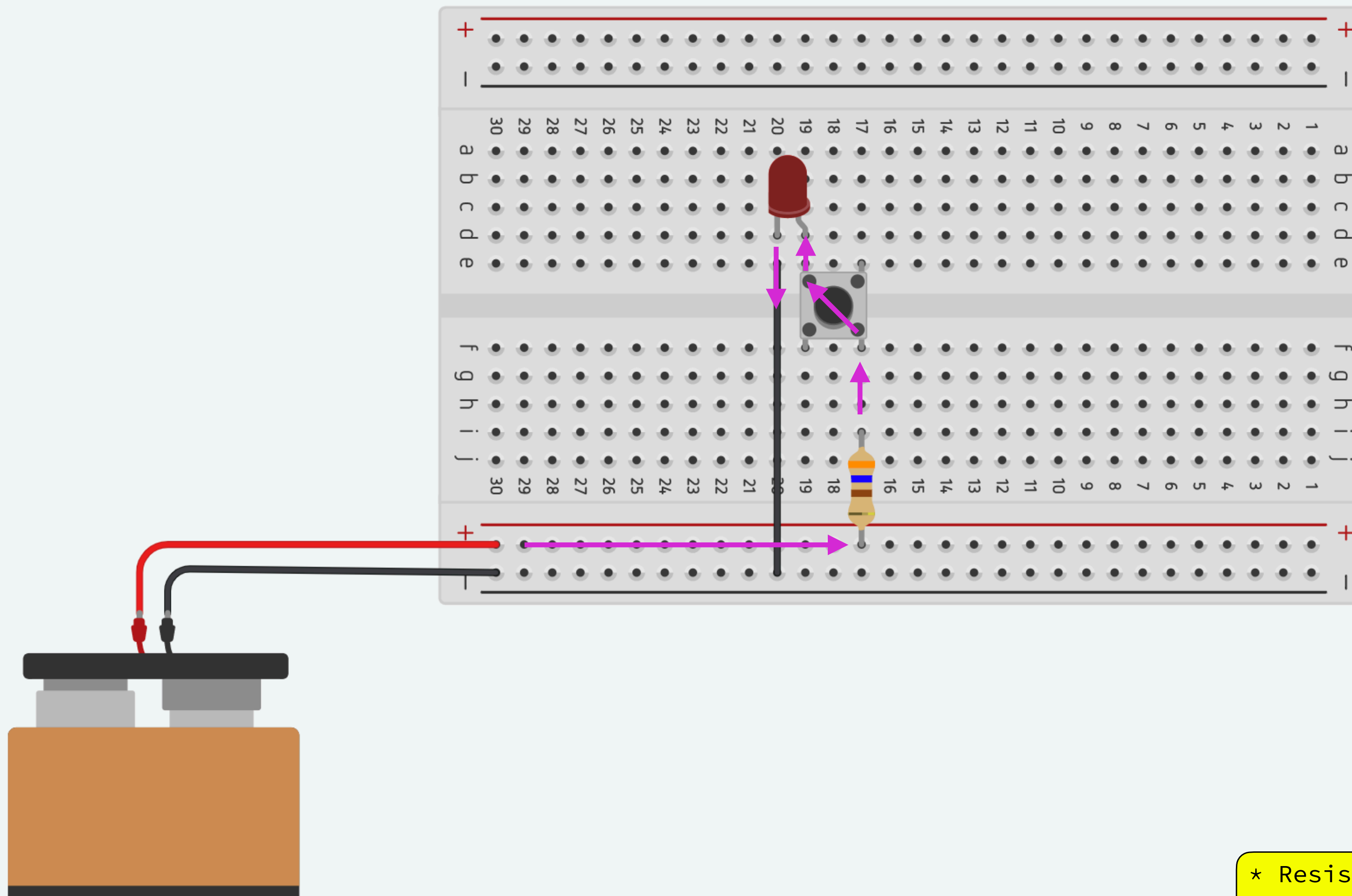
* Resistor is 360Ω (Orange Blue Brown)

Our first LED circuit with a button. The LED will light when the button is pressed.



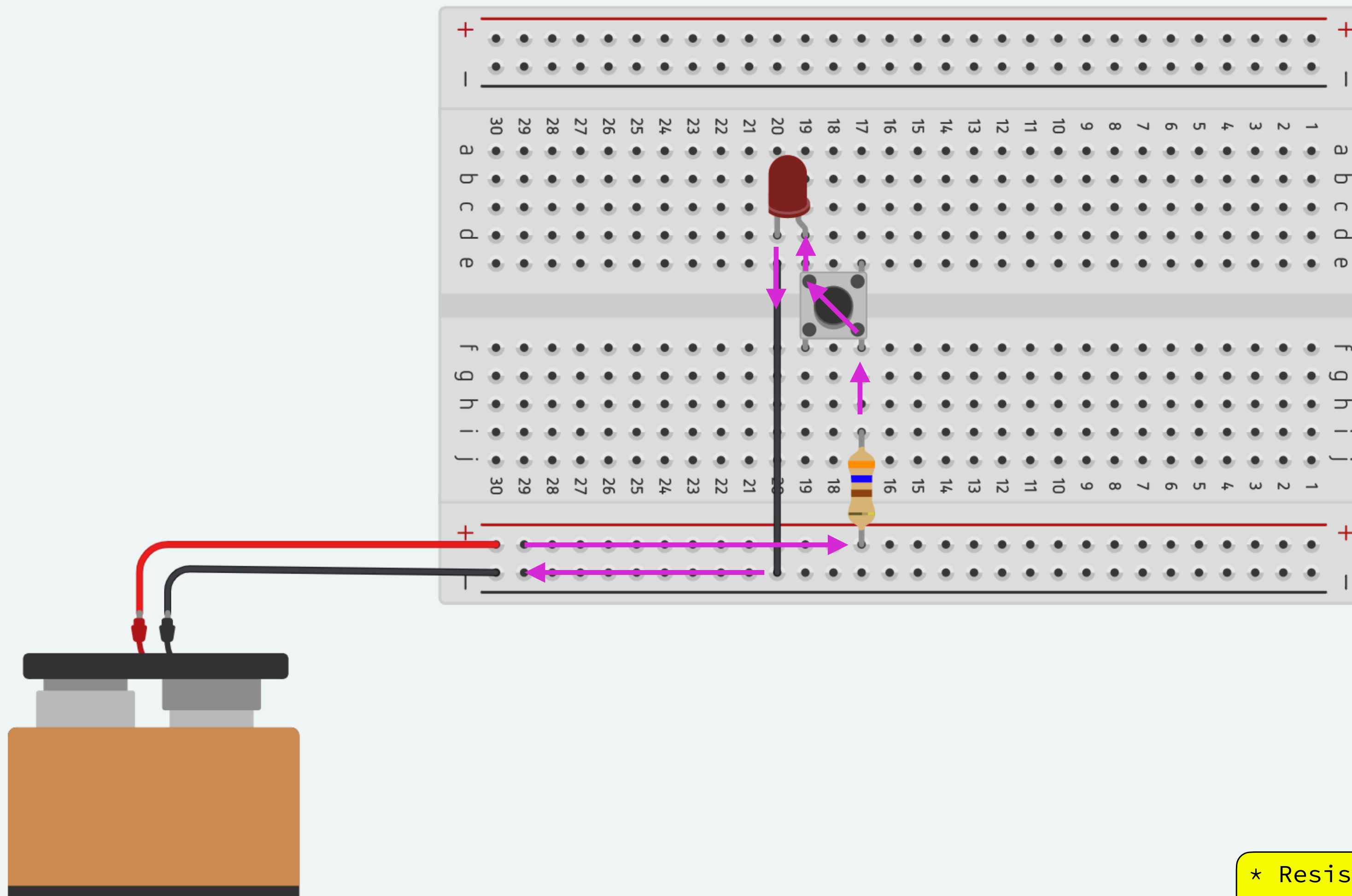
* Resistor is 360Ω (Orange Blue Brown)

Our first LED circuit with a button. The LED will light when the button is pressed.



* Resistor is 360Ω (Orange Blue Brown)

Our first LED circuit with a button. The LED will light when the button is pressed.

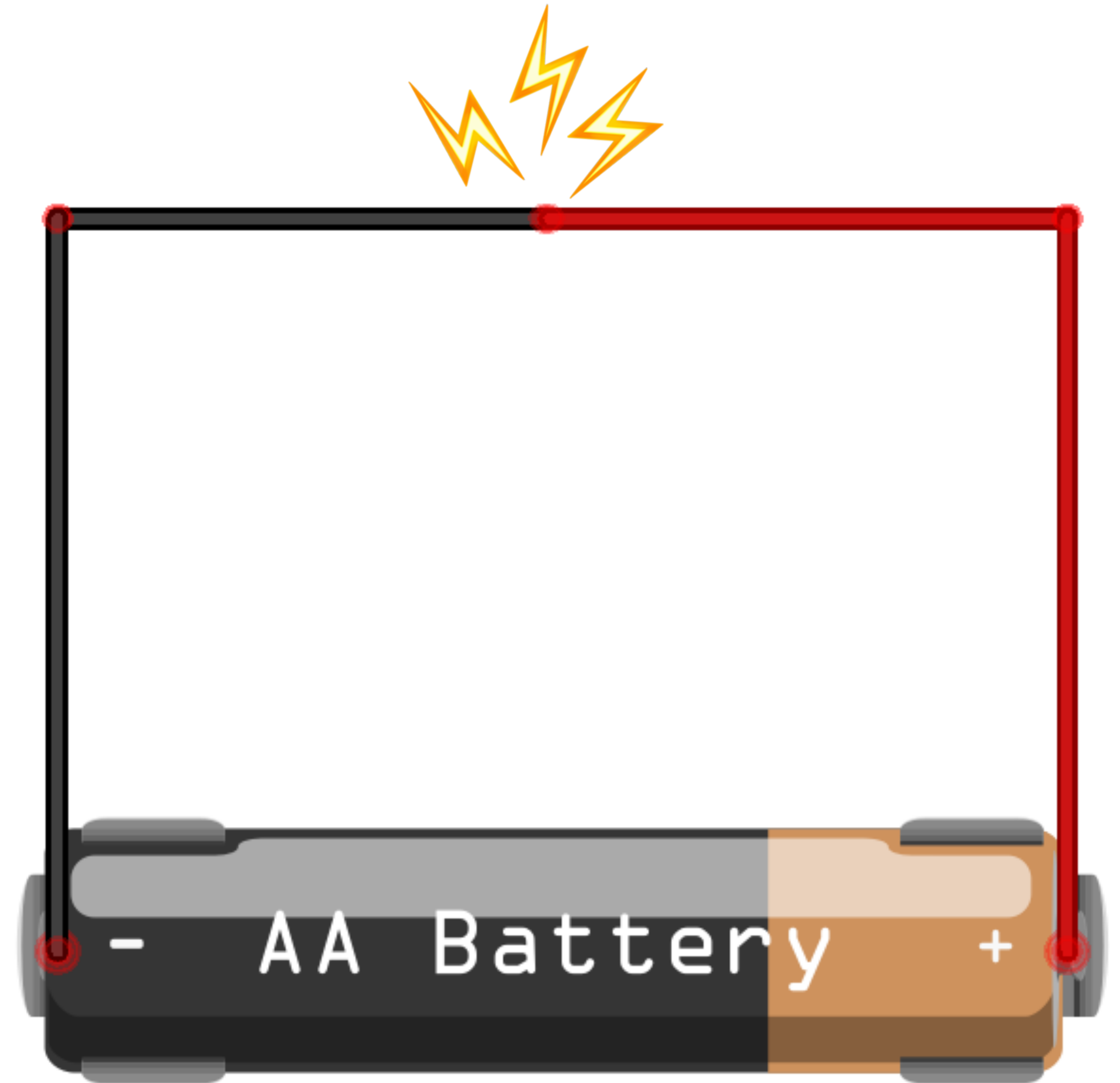


* Resistor is 360Ω (Orange Blue Brown)

Short Circuit!

When the positive and negative wires, instead of being put to work, are connected directly together.

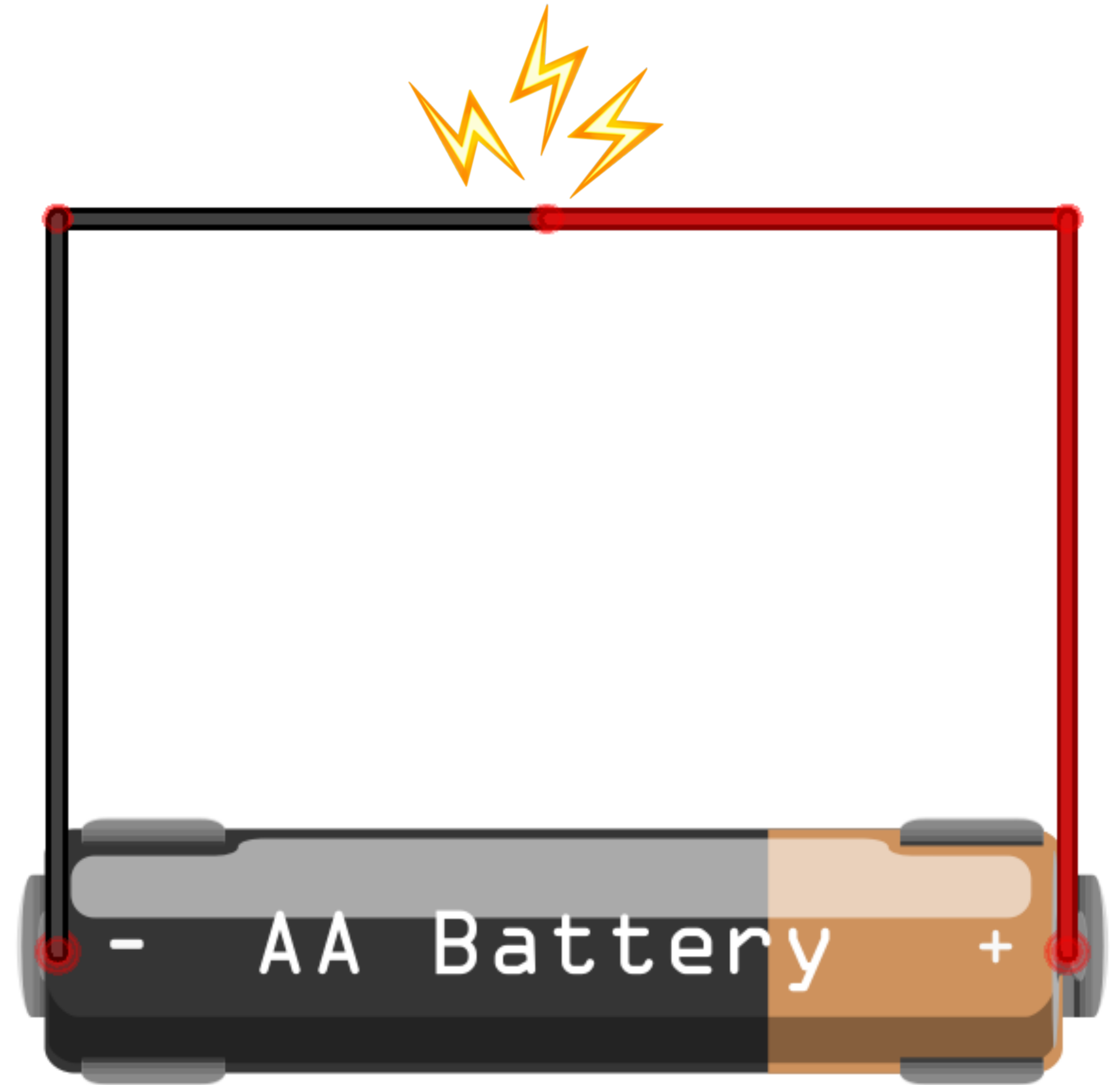
This is bad because the resistance is so low that the current will be very very high which generates heat.



Short Circuit!

This can happen by accidentally connecting the positive and negative terminals of a battery, or, more often, but some loose wires coming out and touching.

Make sure your wires are always trimmed and minimize the amount of metal sticking out.



Try duplicating the button circuit to add more buttons and LEDs.

Experiment with controlling multiple LEDs with one button, or using different buttons to control different colors or patterns.